The Influence of Interactivity on Immersion Within Digital Interactive Narratives

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Abstract: Immersion and interactivity have been argued to be contrasting phenomena's in the sense that interactivity often tends to decrease the immersive experience in many examples of interactive narratives. However, various scholars in the field, such as [2] and [7], believe that this dispute can be reconciled through proper interactive and narrative design considerations. This article, therefore, describes a study where the influence of interactivity on immersion is investigated, with a main intent of discovering a set of fundamental guidelines that would supplement towards creating a basic philosophy for developing successful and meaningful interactive narratives. Following these fundamental theoretical notions, a smart phone-based application of an interactive narrative is designed and implemented with a main intent of reconciling narrative immersion and interactivity in one unified domain. In order to evaluate the validity of the theoretical framework, potential methods for testing immersion in an interactive narrative experience are proposed leading to a development of a methodological framework for testing the hypothesis of this study.

Keywords: Immersion, Interactivity, Narratives, Interactive Narratives, Interactive Storytelling, Narrative Coherence, Narrative Closure, Measuring Immersion.

I. INTRODUCTION AND MOTIVATION

As a form of interactive experience, interactive narratives are believed to have a great potential, of becoming the art form of this century [1]. Since the introduction of the first video games, and the conception of The Holodeck in Star Trek, digital storytellers have been envisioning an idealistic future, where the authors could invite their audience to enter their stories in a form of physically active role, embodying fully immersive and fully interactive experiences. Nevertheless, there are great design and technological challenges involved in such an endeavor, especially because of the first and primary artistic and creative, authorship-based nature of narration that withholds such a potential from being realized. Many interactive experiences offered to the viewer nowadays try to combine various storytelling methods with numerous technological applications and devices that allow viewers to decide which direction the plot of the story should move forward. However, even though such attempts offer novel and sometimes even exciting experiences for the viewer, the usual immersive characteristics of the linear narration tend to become interrupted or even totally disappear.

This tendency is especially often encountered in various interactive storytelling examples offered by computer games and virtual reality (VR) technologies that more and more often allege an apparent discrepancy between narration and interactivity, and thereby minimize the immersive experience for the viewer. Even though most games are highly interactive in such aspects as spatial exploration, collaboration or competition, and physical manipulation of objects, they are not very interactive in terms of story. Most of the games nowadays are either packaged with one linear storyline, allowing only minor variations depending on the player's choices, or present the narratives in a form of pre-rendered cut scenes, constraining the story to always flow in the same manner, independently of what the player chooses. And even though attempts on traversing the issue exist, by offering greater interactivity with the narrative, most often they result in a disturbed immersive experience, losing the resonance that can be found in literature and films. Similarly, even though films, literature and theater, are well known for their immersive – narrative capabilities, many endeavors to involve the audience in an active participatory role often result in analogous, intricate and incoherent experiences.

Such discrepancy is also suggested by many mainstream researches in the field where immersion within a narrative and interactivity are often seen as being two incompatible rather than two coherent qualities, as interactivity is believed to diminish the intrinsic immersive capabilities of narratives. As proposed by many researchers, such as [3] interaction stimulates users' awareness of a medium and their ability to make choices, while during immersion, users are passively engaged and are unaware of the medium. However, other scholars, such as [2] and [7] explicitly state that such a divergence is caused mainly due to the inconsiderate design efforts and thereby could be reconciled through proper narrative and interactive design deliberations.

The controversy, inherent throughout these previously outlined notions therefore proposes an obvious niche for investigation on the influence of interactivity on immersion within digital Narratology. This research, thereby presents a study on various literature within the field, with a main intent of discovering a set of fundamental guidelines that would supplement towards creating a methodological framework for developing an immersive interactive narrative. Subsequently, the study describes how the proposed theoretical framework is deployed in the design and the implementation of the interactive narrative and provides the methodology for conducting an experiment, where the user experiences in interactive and non-interactive scenarios are evaluated, compared and discussed in order to investigate if, and how interactivity influences narrative immersion, and whether their reconciliation in terms of proper design considerations is possible.

II. THEORETICAL INVESTIGATION

To get a deeper understanding of the potential divergence between immersion and interactivity, a theoretical investigation upon these subjects is necessary. This section, therefore, aims to explore general conceptualizations and related research work in terms of the proposed constructs, providing a solid foundation for forming the premises for the development and evaluation of the interactive narrative. As the discussed conflict between the two concepts frames the design of the interactive narrative as the reconciliation of three independent design domains – the immersive, the narrative and the interactive, the theoretical investigation presented in this section is therefore carried out by analyzing these domains separately. Respectively, the first section is dedicated to investigating immersion, with the main purpose of finding implicit characteristics and conceptualizations of the phenomenon and gaining understanding on how immersive narratives can be created. The research reported in the narrative domain section concentrates on examining various aspects regarding narratives, with the main intent on apprehending how conventional narrative theories from literature and film can be applied in an interactive scenario. The interaction domain subsequently focuses on obtaining the necessary comprehension of the fundamental interactive features and gaining an understanding of how interactivity can be incorporated into an immersive narrative. Theories from various perspectives are presented in relation to narrativity in films and games, with the main intent of developing a theoretical foundation for the later development of an interactive narrative experience. The section is therefore concluded with a theoretical framework that provides the underlying criteria for the subsequent design, implementation and research methodology development phases of this study.

1. Immersive Domain

1.1. Immersion and Engaging Experiences

Engaging with media such as computer games, virtual reality, cinema and books has been often argued to award the users with experiences of feeling deep involvement. Given that these phenomena are taking over many media development frontiers, the primary concern of more and more studies is devoted to empirical investigations, conceptualization and quantification of these subjective experiences. Consequently, a wide array of propositions for the terminology exists, leading to a lack of consistency among various media, technological developments and research domains when defining the construct. Throughout prior investigations, a clear tendency towards defining such involving experiences in terms of flow, presence, engagement, or immersion exists, the main dispute often being the degree, the cause of user involvement and the area of media application. Flow, for instance, as introduced by Csikszentmihalyi in [4], characterizes the intrinsically motivated behaviour of artists and sportsmen and is often defined as an optimal experience, tantamount to a Zen-like state. Presence refers to a sense of "*being there*" in a mediated environment and is frequently bound to experiences offered by VR technologies [5]. Whereas, engagement is often a term used to describe user experiences in games, especially at a non-diegetic level – being active in pursuit of a goal, motivated to begin and keep playing [6].

Immersion, on the other hand, in its metaphorical sense is concerned with specific emotional and psychological experiences when being submerged into water, and hence mediated content, a sense of mental absorption, and is frequently encountered in relation to fictional, theatrical and artistic deliberations [2]. Considering the implicit magnitude of the narrative and film domains on this study and the applicability of the term for the intrinsic purposes of this project, the conceptualization of user experience, in terms of immersion, therefore seems to be the most optimal choice among these frameworks and concepts.

One of the most frequently quoted scholars in the field, J. B. Murray transposes the metaphorical meaning of immersion into the context of user experiences with media: *"We seek the* same feeling from a psychologically immersive experience that we do from a plunge in the ocean or swimming pool: the sensation of being surrounded by a completely other reality, as different as water is from air that takes over all of our attention, our whole perceptual apparatus*"* [7, p. 98]. The proposed definition, illustrates the dynamic nature of the immersive experience by means of accentuating not only the psychological dimensions of the phenomenon, but also regarding its technological and sensorial aspects. Such ambiguity leads to differing approaches of discussions about immersion in interactive media. The major discrepancy in the academic research on immersion emerges between the underlying properties that define immersive experiences, as some identify the construct as an attribute of technology and the medium, while others acknowledge it as a deep state of psychological and cognitive involvement, subordinate to the human mind. M. Slater et al., for instance, define immersion in respect to the technological aspects rather than the individual user experiences and describe it as the system's capability to indulge the sense of presence [8]. According to the authors, the level of immersion then entirely depends on the technology's ability to provide sensorial stimuli: *"*the more that a system delivers displays (in all sensory modalities) and tracking that preserves fidelity in relation to their equivalent real-world se*nsory modalities, the more that it is immersive"*. [8, p.1]

Similarly, B. Laurel, in her book Computers as Theater, refers to immersion as a consequence of a sensory stimulus: *"Tight linkage between visual, kinesthetic, and auditory modalities is the* key to the sense of immersion that is created by many computer games, simulations, and virtual-reality systems*"* [9, p. 161]. This technologically driven point of view, offered by both Slater et al. and Laurel, subsequently infers that experiences utilizing more advanced technology and providing a greater stimulus to our senses would always be superior in terms of user experience, hence VR would always be superior to films, which in turn would exceed literature and other forms of art. Following such a perspective within the context of this study, one might argue that film as a medium is comparatively mediocre in terms of immersion. However such conceptualization withholds any further considerations regarding user's involvement with the narrative content provided by the media or the established world of the film, which are essential for the implicit purposes of this investigation. Therefore, a different comprehension of the immersive experience, in terms of user involvement with the mediated content is necessary to be discussed for this research.

Contrary to the technological conception, Ryan proposes that for immersion to take place, an expanse must be offered by a textual world (content), allowing the readers (hence viewers, players and etc.) to accept the mediated reality and temporarily trade it for their physical reality: *"A deep* absorption in the construction/contemplation of the textual world [causing] our immediate surroundings and everyday concerns to *disappear from consciousness"* [2, p. 94]. The essential notion being that the author regards this trade off to be offered not by persuading the viewer's sensorial apparatus, but by deep psychological involvement, based upon the viewer's real-life experiences, cognitive models, and cultural conventions.

Additionally, in accordance to Ryan and in opposition, especially to Slater et al. in [8], K. Brooks also argues that immersion by no means is depended on technology. He states that long before the use of technology immersive experiences have existed in the form of written and oral storytelling and states that adding technology to a mediated content should only compliment the already existing immersion for the person. [10]. He defines immersion as a state, when one is completely drawn into the mediated content for a period of time. The represented world then becomes the substitution of the real world, where the time of the outside world seems to stop [10]. This type of response to narrative relates to a player's absorption in a story as experienced while reading a book or watching a movie.

From the above definitions it is clear, that both Ryan and Brooks refer to immersion, as a deep cognitive and psychological involvement state of change that transforms the human mind from one space to another. M. Green and T. Brock, in their study "The Role of Transportation in the Persuasiveness of Public Narratives" postulate that this change occurs by means of what they term as transportation. In regards to the phenomenon, the authors express: *"we conceived of transportation as a convergent process, where all of the person's mental systems and capacities* become focused on *the events occurring in the narrative"* [25, p. 701]. Following this citation they also propose, that this state of change occurs in terms of how absorbed or transported into a story a reader may become, and how much their real world beliefs change as a result of the story [25]. Although, some aspects of transportation might seem to differ from immersion, in their research, Green and Brock superpose the relation between the two terms: "Transportation can be expressed as immersion or absorption into a *narrative world."*[25, p. 701]. This view on immersion, in terms of transportation, not only coincides with the non-technical perspective of immersion, but also proposes the importance of the narrative context. In the realms of this project, where narratives are an essential part of the overall goal, this approach therefore allows a directly concept-related notion of the phenomena, and provides enough details to further analyze the association between the immersive and narrative domains.

The above reviewed literature clearly exposes the two differing approaches of conceptualizing immersion. However, even though the fundamental aspects of immersion outlined by these two differing perspectives above might seem to differ, one might argue that they rely on unique experiences and different mechanisms that might converge when developing such a versatile experience as an interactive narrative. Therefore, an alternative to the previously discussed notions and perhaps the most adequate conception of immersion in regards to an interactive narrative context can be derived from D. Arsenault's classification of immersion into three distinct categories: sensory, fictional and systemic immersion [11]. The author regards sensory immersion, in accordance to L. Ermi and F. Mäyrä's original proposal in [12], as involvement through our perceptual apparatus, and thus corresponds to the technical perspective of immersion. Applying this notion to a conventional film domain, sensory immersion is often deployed in a form of involving the viewers by stimulating auditory and visual modalities presented through large screen displays and surround sound technologies. Fictional immersion, on the other hand, covers user's involvement with the story in all forms of storytelling, and hence implies narrative immersion through cognitive absorption in the story [11]. This narrative driven standpoint, therefore closely coincides with the previously proposed views of immersion in terms of transportation. Using Ryan's previous conceptualizations, fictional immersion can be described as: *"there is more to* this world [fictional world] that what the text displays of it: a backside to objects, a mind to characters, and time and space extending beyond the display" [2, p. 158]. In film, the reference can be made to the mental maps and hypotheses and meanings that are being formed by the viewer when experiencing the content of a film, hence the mediated story. Systemic immersion, for this project is namely interesting for its participatory properties. According to [11], systemic immersion occurs when some challenges, both physical and mental, are offered to the spectator. In computer games such could be thinking about player's avatar's chances of survival or collection of points to progress to a further level, while in non-participatory forms of media, such as films, systemic immersion occurs while interpreting given clues, solving mysteries to find the culprit before revealed by the narrative.

Arsenault's model of immersion not only settles the debate between the two perspectives of immersion by separating them into distinct categories, but also gives an overview of all possible factors that could supplement towards an immersive experience in an interactive narrative. On the other hand, the model does not imply that all three types of immersion should be present to have an involving experience, while their degrees of emergence can also depend on the specific case [11]. When working within the framework of interactive narratives, such conceptualization becomes useful, as one can argue that all forms of immersion described by Arsenault, to some degree can be present throughout an interactive narrative and should be taken into consideration when designing such an experience for the viewer. However, when considering the overall motivation of this project, and pursuing an investigation on whether interactivity actually has any substantial implications on an immersive experience in a story, sensory and systemic immersion types become of secondary importance, as their implicit conceptualizations do not provide enough resolution for such an inquiry. Fictional and hence, narrative immersion, on the other hand, due to its fundamental properties, might be affected by the clash of immersion and interaction in a unified domain. Such hypothesis is also supported by many scholars, such as, for example, A. Clarke and G. Mitchell in [13] and M. L. Ryan in [2], who believe that when allowing viewer participation in a narrative, it is the sense of engagement with the story that suffers the most.

Following this discussion, it is important to finalize that immersion throughout further deliberations in this paper, will therefore be addressed in respect to its narrative properties and will further be regarded as narrative immersion, hence the viewer's mental absorption and psychological involvement with the story world. Systemic and sensory immersion types, on the other hand, will be of lesser importance for the theoretical framework, however conceptualizations made in this chapter will be used for the later implementation of the interactive narrative experience.

1.2. Narrative Immersion

The following section is dedicated to the concept of narrative immersion and is mainly based on M.L. Ryan's book Narrative as Virtual Reality [2] and largely corresponds to fictional immersion type classified by Arsenault. Even though, Ryan uses the different terms in connection to literature, she also relates the same experiences to watching a movie: "as the reader simulates the story, her mind allegedly becomes the theater of a steady flow of pictures." [2, p. 120] This statement indicates that the experience of watching a movie is relative to that of reading a book as the set of presented symbols and information is understood by the viewer/reader in a similar way and allows a construction of an imaginary world. The main objective with the discussion upon narrative immersion is to understand the key factors and essential elements that when combined with conventional Narratology principles would contribute towards creating an immersive experience for the viewer of an interactive narrative.

Ryan, in her book on the subject, accentuates three main mental processes that constitute narrative immersion and play a determinant role when involving with a fictional world. She divides these into: Spatial, Temporal and Emotional responses to immersion [2]. Spatial immersion, describes an ability of a narrative to involve the reader in mental construction of the spatial setting of the story world and its topography: *"In the most complete forms of spatial immersion, the reader's private landscapes blend with the textual geography. In those moments of* sheer delight, the reader develops an intimate relation to the setting as well as a sense of being present on the scene of the represented events.*"* [2, p. 122]. It is through spatial immersion that the reader of a book can not only grasp the sense of place, and thus imagine the changing landscapes through the eyes of the characters, but also construct a mental model of space, to understand dynamic relations occurring in terms of locality [2]. One can argue that spatial immersion is solely subordinate to literature due to the low degree of the representational methods used and thus the level of imagination required. Nevertheless, when referring to films, even though the spatial conception might depend less on the viewer's disposition, the ability to evoke user's imagination through metaphorical and ambiguous adaptations of space can be used as a prompt for spatial immersion to occur. When referring to spatial immersion in relation to the interactive narratives, Ryan argues that by the use of interactive features it is even possible to improve the apprehension of space in more detail [2]. Additionally, Ryan states: *"The special* power of interactive texts to generate plurality of possible worlds could be regarded as a feature *that facilitates the creation of an immersive plot"* [2, p. 276]. Therefore, when referring back to the main objective of this study, it is assumable that spatial immersion would be the least affected when unifying interactivity and immersion into a single domain. Nevertheless, its importance for creating an immersive narrative remains unquestionable.

The second mental process, as defined by Ryan, is temporal immersion - a type of involvement that manifests through a reader's desire of knowledge that will exhibit itself at the end of narrative time [2]. In conventional terminology, such phenomenon is referred to as suspense, and is greatly deployed especially in literature and films as an easy way to involve the audience with the narrative. In the realms of filmmaking, suspense is known as a dramaturgy technique that makes use of the difference in knowledge between the audience and the characters on the screen. [14]. J. Frome and A. Smuts in [15], distinguish several ways in which suspense is achieved in conventional filmmaking – creating uncertainty, knowledge of the story's outcome and the audience's inability to affect the choices of the character. When working within the realms of interactive narratives, one might argue that the feeling of suspense, and thus our temporal immersion, might suffer from the viewer's ability to change the outcome of the story, due to the fact that "our inability to interact with the fiction increases suspense because we cannot reduce uncertainty by changing the likelihood of the outcome."[15] However, even though in non-linear narratives tension might be minimized by the viewer's impact on the fate of the characters, the factor of uncertainty when choosing the right path among others might compensate this margin. Additionally in [15], the authors suggest that to overcome the uncertainty issue in interactive scenarios, the design should limit interactivity at key points, and thus turn the avatar players into helpless spectators similar to those in film. Moreover, borrowing from game research, such aspects as value of time and putting the player's avatar's life at stake are features that can be utilized for suspense generation [15] and thereby temporally immerse the viewer in an interactive narrative experience.

The last mental process that Ryan uses to define narrative immersion is what she refers to as emotional immersion, or, in other terms, the reader's emotional involvement with the fate of the character [2]. From the early manifestations, literature has proven its ability to elicit a wide spectrum of feelings and emotionally involve the readers/spectators with the characters of the story. In literature this type of immersion occurs when by the narrative means the reader is given an opportunity to gain an insight to the inner workings of a textual character [2]. In the field of interactive narratives, however, many scholars believe that it is in fact the emotional immersion that suffers the most due to the proposed contraposition between immersion and interactivity. Ryan, for instance, states: *"Emotional immersion requires a sense of the inexorable character of fate, of the finality of every event in the character's life, but as Umberto Eco observed in a radio* interview, this outlook is fundamentally incompatible with the multiple threads generated by interactive freedom*"* [2, p. 263]. The author specifically points out that the key to emotional immersion lies behind the fact, that the reader is forced to face the destiny of a character by no means of intervention. From this standpoint, the main issue is that interactivity, by allowing the user to choose the fate of the character and change the course of events, would interfere with emotional immersion. However, when considering the success of computer games, and especially simulation and role playing games, it is often clear that interaction can in fact produce very strong emotional relations with virtual characters, and hence support emotional immersion. Many authors, such as J. Murray in [7], and B. Laurel in [9] believe that to achieve emotional participant investment with characters in interactive stories, it is highly beneficial to give the interactor a first-person role as a character in a narrative. In accordance, game theorists also propose the notion of a player as a protagonist as an essential feature for emotional involvement to exist [16]. Additionally, game research accentuates the crucial importance of establishing the comprehension of the player's role and an understanding that the interactor's actions are non-trivial in order for emotional involvement in interactive stories to occur [16]. Following these notions, the design guidelines should therefore support the notion of emotional immersion within the interactive narrative domain.

Acknowledging narrative immersion through the perspective of spatial, temporal and emotional aspects, as proposed by Ryan, not only provides a suitable basis for the conceptualization of the immersive narrative, but also, allows a better understanding of various methods for the development of an immersive experience and thus can later be utilized as a premise for the design of the interactive narrative. Additionally, delimitations to assess immersion from the narrative perspective, and hence it's three mental processes, provides this research with the basic foundation for the establishment of the methodological framework for evaluating user's immersive experience.

Having these main conceptualizations and fundamental understanding of immersion established for the purposes of this study, it is interesting to continue the theoretical investigation by looking into the narrative domain with the main purpose of forming a solid basis of linear and non-linear Narratology for the later development and evaluation of the immersive narrative, and its reconciliation with interactivity.

2. Narrative Domain

In order to ascertain, how an immersive narrative can be created, an investigation upon the subject of Narratology and narratives seems necessary in this case. The analysis is carried out by firstly apprehending the conventional conceptions of narratives from literary and film domains, in order to discover the essential narrative elements that are integral to a design of a story. Following by a discussion upon various narrative types and structures to form a comprehension of how these narrative elements can be meaningfully organized in a later design of an interactive narrative.

2.1. Definitions and the Narrative Framework

To grasp a general conceptualization of narratives, one can turn to Narratology – the theory of narratives, narrative texts, images, spectacles, events and cultural artifacts that tell a story [17] .H. Potter Abbott, representing a traditional view of many narratologists, defines the term "narrative" as a combination of story and discourse: *"story is an event or a sequence of events (the action), and the narrative discourse is those events as presented"* [H. Potter Abbott, in 18, p. 7]. Following this view, story is the narrative in its virtual form, while narrative is the textual actualization of the story. However, Ryan argues, that this conventional view when defining narratives is asymmetrical, as it is only the story that can be constituted into autonomous terms [18]. Additionally, in [18] the author presupposes that stories are more than sequences of events and refers to them rather as mental images and cognitive manifestations, implying that narrative is not purely a combination between a story and a discourse.

Following the notion, that traditional narrative definitions obstruct the implicit properties of narratives to be defined, in her book, Narratology: Introduction to the Theory of Narrative, M. Bal presents the theory of narratives by establishing a framework consisting of three narrative layers – text, story and fabula. Fabula is regarded as a series of logically and chronologically related series of events that are caused or experienced by actors [17]. Here, the event is a translation between various states, while actors are instances that perform actions at a certain time and place. The author regards text as: "a text in which an agent or subject conveys to an addressee, a story in a *particular medium"*. [17, p. 5] It is the superficial layer of the narrative, the material aspect of the story, such as words on pages in books or the imagery and sound in films. The story is then the content of the text and a presentation of the fabula [17]. The story is the instance that decides the chronology of the events and semantic relations between them, while also distinguishing specific characteristics to actors, time, location or events. Such a proposition, contrary to the traditionalist perspective, allows more resolution when constituting a possible framework for defining narratives, however, the cognitive aspects and mental dimensions of narrative remain unresolved.

Therefore, with an intent to specify the fundamental properties of narrative on both, the semantic and the formal levels, Ryan in [18] defines the narrative framework by organizing the narrative into four categories: spatial, temporal, mental and formal/pragmatic. The first two categories provide an apprehension of space and time for cause and effect to occur. Here the author distinguishes the spatial dimension as the world inhabited by individual existents, while the temporal dimension is regarded as a world that "must be situated in time and undergo significant transformations [that are] caused by nonhabitual physical events" [18, p. 8]. The mental dimension, proposes that the narrative characters should have intelligence, mental life and emotional reactions to the differing states of the world [18]. This dimension, in hand motivates the cause and the effect to occur through agent's actions and reactions to the provided mental stimulus.

The foremost dimension, the formal and pragmatic dimension, suggests that the cause and effect, implying a sequence of events, must form a unified chain that would lead to closure. It is through this dimension that the author refers to the meaningful connections and apprehensions of the story world, as “the story must communicate something meaningful to the recipient” [18, p. 8].

Following such framework to regard narratives, not only allows a general conceptualization of narratives for the purposes of this study, but also, referring to Ryan, following such methodology provides the criteria for determining the text’s degree of narrativity and forms a basis for a semantic typology of narrative texts [18]. Moreover, assuming such a perspective on Narratology in this study also provides the means to directly incorporate the previously proposed conceptualizations of narrative immersion, in terms of temporal, spatial and emotional aspects. Thereof, following the criteria presented in this chapter, in convergence to the immersion theories presented beforehand, should contribute when developing an immersive narrative and can be utilized for later evaluation purposes of this project. Nevertheless, even though the implicit elements of narratives have been identified, it is furthermore necessary to form an understanding of how these can be organized and structured to form a meaningful story. Therefore, the following investigation presents an overview of various narrative types and structures that can later be potentially deployed in a design of an interactive narrative.

2.2. Narrative Types, Structures and Narrative Coherence

To understand the narrative system in terms of linear media, and especially in film, D. Bordwell suggests that it is necessary to distinguish the *“difference* between the story that is represented and the actual represent*ation of it”* [20, p. 49]. Such a definition of a narrative is often regarded in terms of story and plot, where plot is understood as the “everything visibly and audibly present in the film before us” [19, p. 76], the sequence of the events depicted from the story, while story is *“*a set of all the events in a narrative, both the ones explicitly presented and those the viewer infers” [19, p. 76]. The authors argue that the plot is a necessary temporal element that is used to deduce the essential events from the actual story.

In [33] three main narrative plot types are identified: epic, epistemic and dramatic. Epic plots are said to correspond to the archetypal plots of the fairy tales, where the concentration is set on the exploits of a solitary hero and physical actions and human relations that motivate that hero to act [33]. This plot type often deploys the hero’s explorations through the story world in order to reach a goal, conquer a villain, or save another character. The epistemic narrative plot is closely related to the epic type, however, here in the exploration is deployed in terms of solving a mystery. A good exemplification of such a plot type is a classical detective mystery story, where the protagonist is driven by a desire to know [33]. The epistemic plot, among others, is very effective in deploying the reader’s mental properties, in a way that “The intellectual appeal of the mystery story lies in challenging the reader to find the solution before it is given out by the narrative; in order to do so, the reader needs to sort out the clues from the accidental facts, and to submit these clues to logical operations of deduction and induction” [33, p. 7]. The last plot type, distinguished in [33], is dramatic, and refers to the classical Aristotelian definition of dramatic narratives. Here the focus is set on evolving networks of human relations and their progression throughout the story time [33].

According to [2], these plot types are usually organized into a specific narrative structure. Ryan in [2] proposes three types of narrative structures in literature that allow organizing a plot: sequential, casual, and dramatic. Sequential narrative structure covers journals, diaries, or chronicles. This structure organizes the plot as a list of events that can be written down with minimal temporal displacement with respect to their occurrence. The continuity of the sequence is provided by either the permanence in the identity of the main character or a global event chronologically emerging through time [2]. In contrast to the straight-forward sequential narrative, casual narrative structure on the other hand is perceived retrospectively – here the narrator links together events in a casual chain that leads to a specific outcome [2]. Referring to Ryan, a good example of a casual based narrative is a problem-solving scheme, where the sequence of plans is constructed to lead to a given effect that is the goal of the story agent [2]. A clear story goal becomes one of the main characteristics of the structure, and thereby implies that causality in a narrative can be maintained by explicitly defining it throughout the narrative. The last narrative structure type identified by Ryan as dramatic and based on the elusive and variable nature of the story that aims to control emotions and reactions of the spectator in the fictional world [2]. Causality in such a structure is thereby maintained through intrinsic aspects of the narrative context, such as catharsis, self-discovery, empathy and etc. The author points out that dramatic structure often manifests in a closed pattern expansion that consists of exposition, complication, crisis and resolution, also known as the Freytag's triangle [33].

In conventional filmmaking plot organization into a narrative structure juxtaposes to the Freytag's triangle in literature, in terms of creating a narrative arc structure, consisting of *"the setup, the complicating action, the development, and the climax"* [21, p. 28]. Here, the setup is regarded as the establishment of the main character goals and hence the initial situation of the story, the complication, then takes these goals in a new direction by making them more difficult to achieve, while the climax gives premises for accomplishing the goals, and the conclusion allows all obscurities to be resolved [21]. The adaptation of such a narrative arc structure allows the filmmakers not only to maintain the causality in terms of spatial and temporal relation, but also propagates the earlier outlined pragmatic dimensions of narrative.

Bordwell and Thompson in [19] argue that the effectiveness of narrative structures depends on their ability to integrate the cause–and–effect relationship, and therefore is the key element supplying to the viewer's understanding of the overall narrative. In order to accentuate the importance of causality in narrative structures for achieving the narrative comprehension, Thompson points out: "The most basic principle of the Hollywood cinema is that a narrative should consist of a chain of causes and effects that is easy for the spectator to follow*."* [21, p. 10] According to Bordwell and Thompson, that to achieve this, narrative structures should maintain continuity between the scenes, revealing chronology and linear causality [19] and hence support the pragmatic dimension of the narrative and inherit narrative comprehension.

This phenomenon of narrative comprehension in literature is often defined in terms of narrative coherence [2], [19] or narrative closure [22]. B. H. Fiese and A. J. Sameroff provide a good conceptualization of such comprehension in terms of storytelling: "When the narrative is clearly organized and constructed, both the teller and the *listener can assess the "self" expressed within it"* [23]. Narrative coherence, can also be related to Ryan's earlier formal and pragmatic dimension of narrative in [18], as here it is explained that the story "*must lead to closure"*, and *"communicate something meaningful".* In film theory, on the topic of narrative coherence, Bordwell accentuates: *"Comprehending a narrative requires assigning it some coherence [...] the* viewer must grasp character relations, lines of dialogue, relations between shots, and so on" [20, p. 34]. In [2], Ryan proposes that narrative coherence is an essential element for narrative immersion to exist, and thus is an underlying component, when developing an involving narrative experience. Similar viewpoints are asserted by N. Carroll in terms of narrative closure: "a phenomenological feeling of finality, when all the questions silently posed by the narrative are *answered"* [22]. In accordance to previously cited scholars, Carroll agrees that the perspicuous order of temporal and spatial relations and cause–and–effect continuity are the keys to guaranteeing the narrative comprehension in a narrative. However, the author also accentuates the importance of completeness, and a clear ending, that concludes all the questions posed in a narrative by the protagonist, in terms of achieving the sense of closure in a narrative [22]. These notions upon completeness coincide with what is probably considered the basic foundation of narratives proposed by Aristotle in Poetics, where he states that the story should always have a distinct beginning, middle and an ending "which does itself naturally follow from something else, either necessarily or in general, but there is nothing else after it" [Aristotle, cited in 22]. Based on these notions, one can assume that for narrative immersion to exist, narrative coherence and/or closure have to be integral to a narrative structure. Noteworthy to mention, that in this study both terms are identically important, as both provide essential elements that would contribute to an apprehension of a narrative. Therefore, basing on the reviewed literature, the characteristics of both aspects have to be implicit in a design of an interactive narrative, in order for a spectator to conceive a narrative immersive experience.

In most linear media, the linear and uninterrupted narrative structure ensures the means for providing chronological consistency and cause- and- effect relationship between events, and a clear ending, that would hence propagate the narrative comprehension in terms of coherence and closure. Even though, flashbacks and flash-forwards, or jumping through time and space, is possible in both films and literature, allowing inheritance of different spatio-temporal variations into the plot, due to the implied linearity of narrative structures, maintaining continuity between the scenes and maintaining the overall story structure chronological still remains possible [19]. Thereof, in linear media, narrative comprehension is solely dependent on the narrative considerations by the author, rather than being affected by the structure of the storyline. It is in non-linear media, however, that a deep concern is raised in terms of narrative coherence and closure, which in reference to Ryan in [2], then also creates a questionable stance in terms of narrative immersion. In order to apprehend how narrative coherence can be successfully maintained in interactive narratives, an investigation upon how narratives in non-linear forms of media is necessary. In order to grasp such an understanding, our theoretical investigation therefore further continues towards exploring the realms of the interactivity domain, where the comprehension of how the main elements of narratives and immersion can be deployed in an interactive scenario will be formed for the purpose of combining these three distinct domains in a unified design of an immersive interactive narrative experience.

3. Interactive Domain

The previous discussion on immersion and narrativity exposes a clear issue of maintaining narrative coherence in interactive environments, and thereby poses a challenging design problem of an interactive narrative to find a solution for maintaining user's immersion while interacting. Throughout the previous deliberations, suggestions for maintaining narrative coherence in regards to an appropriate design of a narrative structure have been proposed. This chapter analyses this problem further, and provides an investigation of how narrative structure design can be utilized in combination with interactivity, with a main intent of finding a solution for reconciling a cohesive relationship between the narrative structure and the interactive structure in a unified domain of an interactive narrative. This investigation is carried out by firstly apprehending the fundamental aspects and definitions of interactivity and later studying the convergence between interactive, narrative and immersive domains in a context of interactive narratives.

3.1. Defining Interactivity

Interactivity is a key aspect in the context of new media and therefore is also an essential concept when dealing with the realms of interactive narratives. In early research the phenomenon of interactivity is often delimited to the procedural properties offered by the computer technology and the new media. Originating as feedback in Cybernetics in the early 1950's and as the key component of new media in the 1980's, the concept of interaction focused on such aspects as properties and features of the medium/message [34], and user control and participation [35]. However, many later theorists conceptualize interactivity in a more complex manner and try to define its intrinsic properties not only in regards to the properties of the medium. In [36], for instance, the conceptualization of the term is based on qualitative user experiences during interaction. In accordance to such a proposition, Ryan states: "We tend to think of interactivity as a phenomenon made possible by computer technology, but it is a dimension of face-to-face interaction that was shut off by manuscript and print writing and reintroduced into written messages by the electronic medium, together with several other features of oral communication: features such as real-time (synchronous) exchange, spontaneity of expression, and volatility of inscription" [2, p. 204].

When considering this debate, it is possible to notice a major distinction between the technological and the cognitive aspects of the concept. Nevertheless, when designing an interactive experience in a narrative, one can argue that both the technological and the cognitive aspects are equally important and that the distinction between the two conceptions becomes questionable. As an alternative approach when defining interactivity in the context of interactive narration, Zimmerman regards interaction as a combination of four modes: cognitive, functional, explicit and cultural [37]. Accordingly, cognitive is the kind of interactivity that the user can experience when participating with the implicit content of text, functional interactivity is referred to as the user's structural interactions with the textual apparatus, while explicit interactivity allows explicit interactions for the user among the designed choices, and the cultural interactivity occurs outside the boundaries of the text, during the user's reflection upon it. As stated by Zimmerman, these categories are not distinct but rather overlapping and occur in various degrees in all media experiences [37]. In a context of an interactive narrative experience, such interaction modes can be interpreted as constructing relations between the events in the film, choosing among different provided scenes of the narrative structure, interacting with a movie through a remote control and the consequent interactions that happen in post-screening experiences. When creating an interactive narrative, such conceptualization of interactivity seems to be relevant in a sense, that both cognitive and technological aspects are implied in the definition. Such a perspective on the concept therefore implies, that the interaction design in this project should support both, mental, non-participatory forms and the more explicit and functional forms of interaction. The cognitive aspects of interactivity proposed are essential for the overall understanding of the system and thereby the creation of the plot within the mind of the user. Also, incorporating such perspective into this study, allows developing an interactive design that would provide space for narrative immersion to occur. The technological aspects however, are equally important when designing such a construct as an interactive narrative since it is precisely these aspects that make it possible to even consider the more cognitive and representational aspects of interactivity and new media in general.

Ryan appropriately distinguishes two types of interactivity in relation to hypertexts, although extendable to new media in general and terms them as selective and productive interactivity types [2 p. 205]. According to the author, selective interactivity in its simplest form can manifest as random selection among alternatives, such as clicking different hypertexts. By letting readers determine their own path of navigation through a database, hypertexts promote what is usually regarded as a nonlinear mode of reading [2]. Another more purposeful use of selective interactivity is seen in computer games where players are provided with a choice of two different paths - one leading to success and the other to failure, here the game should cue the player to the right or good choice/path. The fullest type of interactivity according to Ryan is the productive one. Regarding this, the author elaborates that the participation and involvement of the user or players' actions are productive - leaving marks on the "textual worlds" by adding objects in the environment or by "writing" the story [2]. Computer games lie in between selective and productive type since the player does not necessarily contribute content, but is involved in a more active participation than only selecting options from a finite menu. This could also be considered in the context of an interactive narrative where a combination of selective and productive interactions could be useful, in the sense that it would provide a sense of freedom and control, allowing the participants to choose which path to take, while additionally writing the story depending on their choices.

Summarizing these propositions, several key points can be emphasized as relevant for further considerations of this project. From the reviewed literature, it is clear, that in a case of an interactive narrative, the interactivity provided to the spectator should deploy both the technological capabilities of the system, allowing an efficient input and output data handling, but also, inherit cognitive features, permitting the spectator to interact with the story not only physically but also mentally, allowing room for immersion to occur. Furthermore, adapting to Ryan's propositions, interactivity should not only allow choosing among the several pre-defined story lines, but also give the spectators a feeling that their actions were non-trivial and contributed to the overall construction of the narrative. Having a fundamental understanding of interactivity formed, it is now possible to investigate its convergence with narrativity and immersion in a unified domain of interactive narratives.

3.2. Interactivity and Narratives

Within the realms of new media and digital technologies, unique qualities of a narrative allow transforming the passive user role into a co – authoritative experience [38]. While, in traditional scenarios, narrative articulation often manifests in the form of receivers constructing and exploiting the stories within their imagination, it is now emphasized that digital media, through the use of interaction, gives the users an ability to act, participate and control the fictional world more actively. Manovich in [39] describes an interactive narrative as follows: the interactor is navigating through a database following links between nodes created by the storyteller. The plot becomes a collection of potential plot events and it then becomes the user's choice to utilize these plots [39]. This process in turn generates a story in the mind of the user. Ryan in [2] agrees with Manovich's basic definition of interactive narratives and states that an interactive text is a "machine fueled by the input of the user" and additionally emphasizes the importance of motivating users to participate in an interactive narrative. Among various reasons the author distinguishes several factors that could motivate the users to submit input in an interactive narrative. One of the distinguished motivational factors is to determine the plot and is known as "choose-your-own-adventure" structure where users select from a list of options and are motivated to uncover the story. Another reason proposed by Ryan is the users desire to shift perspectives of the textual world and thereby follow a different character, situation, or change the plotline. Other motivational factors outlined in [2] are identified as exploring the field of the possible - finding other possible paths or perceiving more of the world, keeping the textual machine going, retrieving data and/or playing games and solving problems. Ryan further argues that a motivation to play games and solve problems is based on the desire of reaching a goal, and therefore states that the design of an interactive narrative should inherit a narrative goal, that would motivate the spectator’s interactions. The author suggests that a possibility could be to present a goal in a form of a mystery, which has to be solved for the narrative to continue. When referring back to the earlier discussions, it is evident that epistemic plot of a casual narrative structure, due to their implicit goal-based and problem-solving schemas would appropriately suit this proposition and allow an easy incorporation of a narrative goal into an interactive structure.

In accordance to Ryan’s emphasis on the importance of narrative goals D. Mott in [16] states: “Interactive stories can only succeed when the author is able to willingly give up sufficient control, such that the audience has the ability to contribute meaningfully. Authors should try to stop thinking about story as a connected series of known events and concentrate on the goal of the story. While the details of the story make it come alive, it is the underlying moral values and the driving goals embedded within the story that truly makes it powerful“[16, p. 121]. Here, Mott accentuates that the author can set up goals for each scene and each event, which can fulfill the different parts of the narrative structure, but it is the interactor that should contribute meaningfully towards reaching the narrative goals. Therefore, it is important for the interactive story to provide mental space for the user, and allow one’s own intellectual capabilities to contribute towards reaching a goal [16]. Mott further deliberates, that in order to do this, it is firstly significant to allow the user a clear understanding of his role and identity in a story: is one a director, an actor, a character, or an outside player who controls a character? Understanding how his role is affecting the narrative and how one can impact the story, gives the participant the power to make decisions that allows reaching a narrative goal and creates the plot of the story. Moreover, referring back to the discussions upon emotional immersion, presented earlier in this research, it is evident that by allowing a user a clear understanding of his role that emotional immersion can be achieved. Additionally, by incorporating this aspect in a design of an interactive narrative, the sense of productive interactivity can be produced, where by understanding his role in the story the interactor would be able to deploy meaningful and non-trivial actions.

Another aspect to consider when designing an interactive narrative is that of interactivity’s compatibility with the traditional narrative types and structures. In [2], it is argued that the compatibility of interactivity and narrativity depends on how extensively the narrative is defined. From the previous discussion upon narrative types, three types of narrative were distinguished: the epic, the epistemic and the dramatic. Ryan in [33] deliberates upon how these narrative types coincide with interactivity. Here the author proposes that, both the epic and the epistemic plot-types are mutually compatible with interactivity and are commonly found in adventure, shooter and MMORPG games. The goal-oriented action found in games closely coincides with the intrinsic definitions of epic and epistemic narrative plot types and therefore support user interaction [33]. These narrative types take advantages of visual recourses of digital systems, providing the interactor a set of clues and objects and a spatial setting to explore and encounter them. Furthermore, both epic and epistemic plots are fully compatible with the conventional game controls that allow exploration, picking up objects and examining clues. On the other hand, the implementation of an interactive dramatic narrative type, according to Ryan, raises countless problems, mainly due to its emphasis on the evolution of interpersonal relations and the incompatibility of existing technology to actualize it [33].

A similar tendency exists when incorporating interactivity with the basic narrative structures, distinguished by Ryan in [2] and discussed earlier, namely sequential, casual and dramatic. Here it is discussed, that the sequential and casual narrative structures are mutually compatible with interactivity, while the dramatic structure lacks its consistency in interactive scenarios. Combined with interactivity, the sequential narrative structure can evolve for instance into a form of linked collection of lexias that refer to the same individual and incorporate self-contained events [2]. Nevertheless, the author proposes the combination between the sequential narrative structure and interactivity rather boring, and lacking the characteristics of a rewarding interactive experience [2]. The casual narrative structure, on the other hand, is regarded as both, interactively-applicable and rewarding. The purpose of interactivity within casual structures is to discover the plan, set up by the system, and to overcome the hurdles, arranged along the way. Taking a wrong turn or failing a test does not ruin narrative continuity, because this framework allows both stories of success and stories of failure [2]. The applicability of this narrative structure in interactive scenarios, therefore relates very well to the previously discussed, Mott's arguments in [16] of setting a goal for the story providing room for the user to create the plot. Additionally this allows for both selective and productive interactivity to be utilized since having an explicit goal must depend on the options provided for it to be achieved and thereby the contribution to the supplying of the story depending on the choices . The last narrative structure type, the dramatic, is on the other hand proposed to be hardly compatible with interactivity as the implementation of such a narrative structure requires a delicate coordination of the user's actions with the goal of the system [2]. All choices given to the user constitute a potential threat to the global design, and thereby the quality of the overall experience. As a designer you must be able to foresee all possible actions of the user, given the effect that the user feels that he determines the plot [2]. The obvious complexity of the dramatic narrative is argued by Ryan to be the reason why the dramatic narrative structure is the most problematic to implement in an interactive narrative.

As seen from these deliberations upon narrative structures and their relations to interactivity, it is evident that some are more applicable than others to interactive scenarios in terms of narrative coherence. Recalling the previous investigation in the narrative chapter, a questionable stance regarding the relation between interactivity and narrative coherence was created, due to the difficulty of maintaining the implicit properties of narration in terms cause-and-effect relationship when combining narration and interactivity. In regards to narratives in non-linear forms of media, Ryan states: *"Interactive Narratology [...] involves the same building blocks as the traditional* brand: time, space, characters, and events. But these elements will acquire new features and *display new behaviours in interactive environments"* [18, p. 100] However, maintaining the cause–and–effect relationship in non-linear forms of media is often a difficult task. Therefore, many interactive narratives fail to preserve a clear link between temporal and spatial relations between the events and thus easily fail to provide a viewer with an immersive story experience.

The cause–and–effect relationship often highly depends on the level of interactivity and the chosen story structure, or story architecture, as referred to by Ryan [18]. In less interactive environments, the plot is offered to the viewer in a similar way as in linear narrative structures however, multiple linear stories can exist as a consequence of added interactivity. In such structures, even though the overall architecture of the narrative is non-linear and offers many story lines, the spectator only apprehends one story line per viewing [24], thus maintaining the cause – and - effect coherence consistent, and thereby enabling the means for providing narrative closure.

Regarding narrative in highly interactive scenarios, such as hypertext, computer games and virtual reality, causality can be much more affected compared to those in linear forms of media. As such scenarios provide the viewer with much choice and interactive freedom, thereby they make maintaining causal, temporal and spatial narrative elements in a logical and a coherent way, problematic. As outlined in [24], in highly interactive scenarios, the issue arises due to the fact that the author of the narrative has no control over the actions of the interactor, and cannot foresee every possible narrative pathway, which hence leads to combinatorial explosion of possible storylines. Due to this, it becomes challenging to guarantee the causal connections and semantic relations between events to be consistent, and thereby to achieve narrative coherence and closure. According to Ryan, immersion and narrative coherence can be reconciled through the cohesive relationship between narrative structure and an interactive structure [2]. Moreover, the author proposes that non-linear narrative structures should borrow aspects from the linear narrative forms, which would in hand lead towards the establishment of not only coherence and closure in narratives, but would also provide the basis to maintaining narrative immersion easier: "the *development of interactive texts [...] depends on its ability to create an immersive experience, and* classic narrative structures are the most time-*tested recipe for keeping the user spellbound"* [2, pp. 243 - 244]. Following this notion and the previous considerations upon the traditional narrative structures, one can propose that the interactive narrative should deploy the same narrative structure principles as those used for linear storytelling. If such a narrative structure is maintained in a non-linear context, with an appropriate design of an interactive structure, narrative coherence would then be ensured, disregarding the implied non-linearity of the medium.

Having settled the main aspects regarding interactive narratives and investigated the relation between interactivity and narrative structures, and, the next step of this investigation is to find out how is it possible to deploy an interactive structure maintaining a coherent narrative. For this purpose the following investigation discusses various models of interactive structures that would provide towards a design of an interactive structure for this study.

3.3. Interactive Structures

Ryan argues that the potential and the success of interactive narratives depend on its architecture - its system of links between nodes [2]. She proposes and exemplifies nine different models of interactive structures. The distinguished models vary according to the amount of interactivity and degrees of freedom implied in their structure, and also in their potential to afford narrative coherence while interacting. Such models, as The Complete Graph, The Network, The Maze and The Action Space, Epic Wandering and Story-World, imply very large degrees of interactivity and link very large amounts of nodes to each other. These structures are said to be represent interactivity of especially complex environments and can hardly maintain narrative coherence throughout the global level. Therefore, the applicability of these interactive structures when creating interactive narratives is questionable, and is more suitable when designing online networks, hypertext and large game databases.

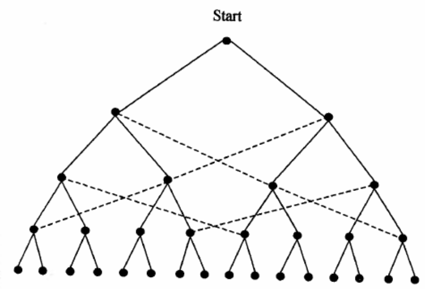

Figure 1 - Tree Structure, retrieved from [2]

A Tree structure (see figure 1), on the other hand, ensures narrative coherence and therefore is more applicable when designing an interactive narrative. Even though, the overall architecture of such interactive structure is non-linear, the tree structure can be seen as a combination of multiple structures. As it allows no circuits, no possible return to the decision point and provides only one way to reach a given terminal node, the interactor can only apprehend one storyline per session [2]. A linear itinerary from start node to end node therefore guarantees a well formed and coherent story [2]. These tree structures are particularly efficient when modelling decision points in a narrative, and therefore supply selective interactivity. However, the problem regarding this kind of structure is the amount of possible events to be created as Ryan argues “it runs into a combinatorial explosion” [2, p. 248] which for instance takes sixteen different plots, with thirty-one different fragments in order to make four decision points. Due to such an exponential growth of an interactive narrative, therefore, it becomes almost impossible to pre-plan all the possible branches beforehand. Even though, it becomes too complex to design the entire interactive narrative based on this structure, the branching technique deployed by this example, could be used at limited points of the story, where the choice of the interactor is significant.

The Directed Network structure (see figure 2), on the other hand, is argued by Ryan to offer “a more efficient management of choice, because the strands of plot are allowed to merge, thereby limiting the proliferation of *branches”* [2, p. 252]. This type of structure can be thought of as a way to manage the expansive nature of the tree structure described above. Here, the vertical axis represents different ways to reach a point, while the horizontal axis inherits a spatio-temporal representation [18]. Because of such an organization of the interactive structure, the chronological consistency and causality are maintained, but at the same time the interactor is granted some freedom. However, due to the effect of early choices on the later ones, it is noteworthy to keep cautious that the intellectual interactivity of the user and the design of the narrative are crucial to match in order not to diminish the expectations of the interactor.

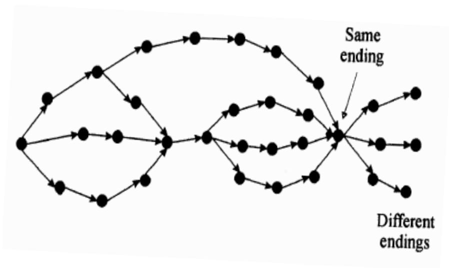

Figure 2 – The Directed Network structure retrieved from [2]

The Vector with Side Branches (see figure 3) interactive structure allows telling a determined story in a chronological order [2]. The structure of links enables the user to take short side trips to “roadside attractions” to gain more narrative detail. This type of structure maintains narrative coherence by its chronological order but provides limited interactivity [2]. Nevertheless, in an interactive narrative the purpose of the interactivity could be to discover background stories or details of different characters. The issue regarding this type however is the possibility of losing focus on the main story while navigating through the side plot events and the fact that the participant can choose not to interact leaving the narrative solely linear.

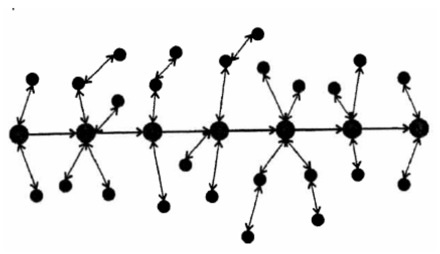

Figure 3 - Vector with Side Branches structure retrieved from [2]

Another structure distinguished by Ryan is The Hidden Story (see figure 4) which is argued to suit mystery stories and computer games with the main intent of discovering the

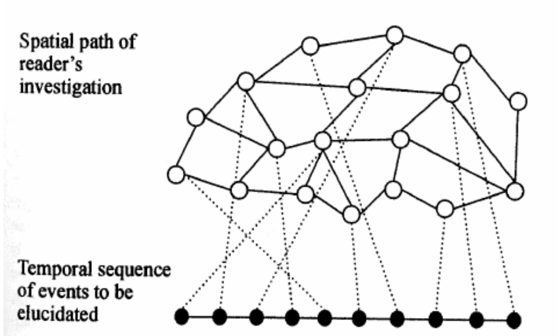

Figure 4 - Hidden Story Structure from [2]

pre-history of the story/game world [18]. It provides two narrative levels; the spatial path of the users investigation at the top level – an atemporal network of choices that determines the path of investigation and the temporal sequence of events to be elucidated at the bottom level [2].
The plot is then formed on behalf of the choices made in the attempt to reconstitute the underlying story. This architecture is argued by Ryan to be the only one that places interactivity in the service of narrative desire since the reader's action is to discover rather than create, and because the story to be investigated is itself unilinear, determinate, and external to the interactive machinery.

Regarding the previously stated notions from Ryan in [2] and Mott in [16] upon the significance of maintaining narrative goals in interactive structures, it is evident that The Hidden Story, The Directed Network and The Vector with Side Branches structures imply such a proposition. However, among the three, The Hidden Story structure provides most freedom of interpretation and thus most room for cognitive levels of interactivity. Additionally, as investigation is implicit to its construction and provides mystery, The Hidden Story structure, allows deploying intrinsic motivational factors for interaction, and coincides with the previous arguments made by Ryan. Nevertheless all of the structures guarantee narrative coherence in terms of chronological consistency and clear cause-and-effect relationships between events, and thereby are applicable to an interactive narrative scenario. Also, all three of these structures allow deploying both selective and productive types of interactivity which therefore provide interaction in terms of user choices and the sense of non-trivial actions that affect the narrative. Nevertheless, the choice of a specific model, or a combination of few, is mostly dependent on the choice of a narrative structure and a narrative plot type, and therefore will be adapted in more detail to a specific scenario in the design of an interactive narrative.

3.4. Immersion and Interactive Narratives

From the propositions made in the previous chapters general considerations regarding interactivity were made, and a relationship between interactivity and narratives was established. Having the most important elements of interactivity identified and the main requirements for interactive narrative structure settled, it is now possible to discuss the stance of interactive narratives in terms of immersion.

According to Bolter and Grusin in [3], one issue that emerges due to the convergence of these domains can be conceptualized in terms of immediacy and hypermediacy. Referring to the authors in regards to immediacy, it is proposed that during immersion the interface becomes transparent and the users develop a relationship with the contents presented by the medium, rather than being aware of confronting the medium. On the other hand, hypermediacy is presented as a contrary term that implies interactivity, during which the users become aware of the medium and the provided interface, and therefore concentrate their full attention towards the medium itself rather than the mediated content [3]. Following these notions the structure and design of the interface therefore becomes essential components of an interactive narrative that supports an immersive experience. In terms of an interface design, Bolter and Grusin in [3] propose - the less transparent and immediate the medium is the more hypermediated the experience will be. In the realms of an interactive narrative, where the clash of immediation and hypermediation is possible to occur, therefore the author's propose that maintaining an interface that is transparent, but still supporting interactions through a hypermediated environment, should lead to an immersive interactive experience [3]. It is important to denote, that a transparent interface in this context should be understood in its literary sense, as it denotes the absence of buttons, menus and hyperlinks that are usually implicit to hypermediated environments: "interface should disap*pear [...], it must not* *intrude into the user's consciousness as he or she wor*ks with a computer application; The user should look through the interface, rather than at the interface" [3, p. 18]. The examples of these types of interfaces are commonly found in VR applications, and FPS games, especially in terms of controlling the character avatar with juxtaposing body movements.

Regarding the interface design for interactive narratives, Ryan additionally proposes that in order to support an immersive experience, the interface has to be natural: "Ideally, users should interact with computer-generated worlds in exactly the same way they interact with the real world: through language and the body"[33, p. 8]. As such a notion is still technologically hard to realize to the fullest, an alternative is to use symbolic performance of physical actions through the provided controls. Furthermore, Murray, in accordance to Ryan, states that "the screen itself is a reassuring fourth wall, and the controller is the threshold object that takes you in and leads you out of the experience" [7, p. 108]. This extends the notion of a natural interface, in terms of the body, integrated into this virtual realm as the controller. The produced actions then become integrated with the story world and control is very closely tied to an object in the fictional world, such as a screen cursor that turns into a hand [7]. This in regards to new media occurs in VR systems and games when the interaction becomes a part of a first person activity. Referring to the above cited authors, it is possible to postulate that maintaining an immersive experience in an interactive narrative highly depends on the interface design. Following these notions that the interface in the context of an interactive narrative is the main tool affording user's interactions and creating meaning with the narrative structure, we suggest that it becomes a component that ties immersion and interactivity together. Therefore, in accordance with the cited authors, an interface that is transparent, natural, and deploying first-person activity, would supplement towards creating not only an interactive, but also an immersive experience.

Furthermore referring to immersion in terms of an interactive narrative, Murray in [7] suggests that within the range of effort required for immersion, a clear boundary between narrative and interactivity applied is important to have in mind when trying to immerse a viewer in such experiences. She further argues that if gameplay consumes most of the available cognitive resources, little room will be left for perceiving complex narrative patterns, and little space for adding to immersion. Conversely, focusing on the development of the sense of narrative reduces the player's need and capacity for a highly engaging gameplay design. Even though the author regards the enjoyment of immersion as a participatory activity, she also states that "a stirring narrative in any media can be experienced as a virtual reality because our brains are programmed to tune into stories with an intensity that can eliminate the world around us." [7 pp. 98-99] This approach suggests and supports participatory activity in relation to narratives however maintaining it on a low level by neglecting an incorporation of complex gameplay design. Following these notions it is therefore evident that considering an appropriate relation between the amounts of interactivity and narrativity is necessary to ensure an immersive experience in interactive narratives. Moreover, Murray proposes that having a lot of information both in terms of interaction and narrative context, can be overwhelming and thereby ruin narrative coherence [7]. Hence it becomes important to limit the amount of presented information in terms of both, interactivity and narrativity, in order to ensure the user's apprehension of the overall structure of the interactive narrative.

Manovich in [39], additionally accepts the relationship between interactivity and narration and/or immersion proposed by Murray, and argues for their cooperation and further agrees in the statement provided by Murray, that in order to create successful interactive experiences viewers should be allowed to oscillate between a state of immediacy and hypermediacy and hence interactivity and narrativity [39]. This, according to Manovich in [39], an immersive interactive design should allow the user "to oscillate between the roles of viewer and user, shifting between perceiving and acting, between following the story and actively participating in it" [39, p. 30]. By deploying this concept in a narrative design, both the immersive nature of the story and the involving moments of interaction are therefore able to be reconciled.

Furthermore on the topic, Ryan in [33], states that: "In real life, we interact with people and the world on a fairly constant basis, though there are moments when we are stripped of agency and forced to watch as spectators the events that determine our destiny. To reproduce this aspect of life, interactive narrative should make interactive moments the rule and passive moments the exception, rather than limiting agency to a few decision points separated by long stretches of passive watching" [33, p. 8] This therefore extends the notion of oscillation proposed by Manovich, in terms that, even though the shift between immediacy and hypermediacy is available, the phase of hypermediacy should be more explicitly operated. This would allow offering the interactors the previously proposed shift between roles, but at the same time would ensure that their actions in an interactive narrative are not always constrained.

The literature presented here proposed that maintaining an interface that is transparent, natural, and deploying a first-person activity, in combination by supporting interactions through a hypermediated environment affording user oscillation, should lead to an immersive interactive experience. These aspects of interactivity in a narrative are proposed as the leading issues that play the determining role when constituting the success of an interactive narrative and gives reason to believe that their integration would supplement towards creating an immersive experience for the user.

3.5. Interactive Medium

The previous assumptions made throughout the theoretical research of the Interactive domain, proposed that both cognitive and technological aspects should be supported by the design of an interactive structure. Furthermore, it was stated that the technological aspects are essential as they provide the means for the cognitive and representational aspects of interactivity and, in general, new media to be possible. Subsequently, the interface was suggested to be the key component that ties immersion and interactivity together in the realms of interactive narrativity. Additionally, adapting D. Arsenault's perspective to view immersion in terms of its sensorial, fictional and systemic properties in [11], it was proposed that even though fictional immersion is of most interest in interactive narratives, sensory and systemic types of immersion are required to support the overall experience. As these aspects extensively depend on the choice of interactive media, a suitable medium type, supporting both immersion and interactivity for such an endeavour has to be chosen. Therefore, in this chapter the focus is set on investigating various media platforms suitable for presenting the content of an interactive narrative medium and will furthermore provide examples of previous work done in relation to the respective mediums.

One of the most obvious medium types supporting narrativity is the literary domain. Nevertheless, even though interactive books are possible for deploying interactive narratives, the amount of interactivity is limited to turning to different sections of the book. Evidently, this type of medium only coincides with the selective interactivity, and not the productive one, and thereby is not relevant for further considerations in this project. Similar limitations are offered by the film medium, since one of the only possible ways to afford interactions is by the means of the remote control. On the other hand, due to the implicit computational recourses, and the support of wide range of available interaction controls, computer based devices, promote both selective and productive interactivity types, and thereby are of more relevance for the development of an interactive narrative for this study. To support productive interactivity, in the field of computer generated experiences various mediums exist and are in this section divided into Virtual Reality systems, Computers, and Hand Held Devices.

Virtual Reality is a medium that provides representations and simulations of 3D environments, enabling user interactions with a generated reality, often by means of Head Mounted Displays and motion sensors. VR’s are often used to simulate real world situations offering training - for situations that are too dangerous or expensive in the real world or treatments. Despite the fact that VR systems afford possibilities of creating immersive sensory and systemic experiences, it is rare that narrative elements are incorporated as part of the experience. Additionally, in relation to this specific study it is possible to point out several issues regarding narratives in VR. Firstly, most available VR systems nowadays are limited to a specific location, are expensive, and often use intrusive equipment which automatically makes the user experience less casual. It might even be possible to argue that this could be part of the reason why the examples of interactive narratives within VR’s are so rare.

Usual computer mediums, such as video games, on the other hand seem to be the most common choice for presenting interactive narratives so far and are also argued by Murray to be an affording medium when combining immersive and interactive experiences [7]. Several successful attempts in developing computer-based interactive narratives can be therefore outlined further.

The Last Express was an early attempt to push the boundaries of interactive narratives within the field of video games. It is categorized as an adventure game, where the narrative sequence is dependent primarily on real-time flow, as all characters act in accordance to real-time. Therefore, due to the real-time narrative flow, a player can only encounter some events and not others and thereby skip narratively essential aspects. This proposes a questionable stance in terms of the overall understanding of the story, and might therefore disrupt narrative coherence. However, the elapsing time creates suspense and leaves no time to think too much forcing the participants to act fast.

Façade, a video game, based on the novels of Virginia Wolf, is a later attempt to create an interactive story often referred to in various studies on interactive narratives [16]. The users play a good friend of a feuding married couple with a fragile relationship. The story begins when the player visits for a dinner and overhears the end of a discussion. The purpose of the interactivity is not to collect items and it is not even necessary to preserve the dissolved marriage. It provides different paths for the participant with the attempt to create a satisfying dramatic narrative. The participant is free to explore all paths of interest and there is no way to win or lose. The story however, encourages the participant to come back and play to discover new outcomes or details of the complex dramatic narrative. The interactivity in the game is provided by the user typing text commands to interact with the couple. This interactive narrative distinguishes itself from other examples in the field in terms that the interaction possibilities provided are dialogue-based and are not based on opportunity to choose between different pre-rendered scenes. Even though this interactivity type provides an exciting experience and opens new possibilities, the narrative design is highly limited to the technological capabilities of the system, as an extensive database has to be created for recognizing the text-based input provided by the user. Both Façade and the Last Express are early attempts to create such interactive narratives and are therefore solely computer based experiences.

With the vast development of technology these concepts of interactive narratives have broadened to hand held devices, and specifically to applications for Smartphone’s. One of the latest attempts in Smartphone based interactive narratives The Witness, a combination of innovative film making and augmented reality, where the participant becomes a part of the real-life game. The interactivity is possible in terms of gathered data based on location and inquiring the experience through the screen of an iPhone. The use of Smartphone and augmented reality combined with additional technologies such as real-time interactions, social-networking, and game mechanics provides an exciting innovative experience within the field. Because of the convergence of various Medias, and an innovative domain in terms of interactive narratives, the use of Smartphone's therefore seems as an interesting choice to consider. Due to the vast development, Smartphone applications have become very popular throughout the recent years, and are eligible to support sensory and systemic immersion types, and provide productive and selective interactivity just as effectively as computers. Furthermore, interfaces provided by the smart-phones, closely coincide with the transparent interface notions provided by Bolter and Grusin in [3]. These reasons thereby provide a good stance in terms of choosing a smart-phone as a medium for developing an interactive narrative in this study.

III. THEORETICAL FRAMEWORK

The conducted theoretical research not only provides a basic understanding towards conceptualizing immersion, interactivity and narratives, but also allows a better apprehension of the fundamental aspects and their convergence in the realms of interactive storytelling. Based on the knowledge gained from the previously reviewed literature in the immersive, narrative and interactive domains, a set of fundamental guidelines that would supplement the creation of a basic philosophy for the development of an interactive narrative can thereby be proposed.

Our theoretical framework for developing an interactive narrative begins with a comprehension of narrative immersion, and especially with an adaptation of Ryan's proposed theory to view immersion in regards to its emotional, spatial and temporal properties [2]. Based on these notions criteria for sustaining immersion in interactive narratives can be postulated further:

1) <u>Spatial Immersion:</u>
   - Spatial design should evoke imagination.
   - The design should allow a mental construction of the spatial setting.
   - The design should indulge exploration through the narrative world.
2) <u>Temporal Immersion:</u>
   - The design should evoke spectator's desire of knowledge.
   - The design should create suspense when choosing among paths of the story.
   - The design at times should employ the user's inability to affect the choices of the character.
3) <u>Emotional Immersion:</u>
   - The design should allow gaining an insight on the main characters.
   - The design should clearly define roles of characters.
   - Provide comprehension that the interactor's actions are non-trivial.
   - The design should deploy the notion of the interactor as a protagonist.

From the literature review in the narrative domain, and especially from the propositions by various scholars, such as [2], [17], [18], and [19], certain elements of narrative can be identified as significant for the development of an immersive narrative for this study. Based on these notions, the criteria for the narrative design include:

1) Characters: Intelligent agents, that allow emotional involvement, have mental life and reactions to the surrounding world.
2) Events: Should represent various states of significant action for actors to perform.
3) Space: The design of narrative space should allow room for events to take place and for actors to perform actions.
4) Time: Should provide chronological consistency of events and semantic relations between them, create suspense.
5) Context: Should have an overall theme, provide mental stimulation for the user and allow making meaningful connections.
6) Narrative structure: An organization of a narrative into a structure, allowing the introduction, complication, development, climax and conclusion. Should maintain a clearly defined story purpose, and goal.

Moreover, through earlier deliberations on narratives, we have proposed narrative coherence, as a significantly important aspect that is crucial to be sustained in order for narrative immersion to occur. Adapting the notions from previously cited scholars, such as [2], [19], [20], [22] and [23] certain aspects that would support maintaining coherence in interactive narratives can be distinguished:

1) Causality: all narrative elements should imply a causal and meaningful relation to one another.
2) Chronological consistency: The narrative structure should have a concrete beginning, middle and ending, that deploy chronological and semantic relations between events.
3) Completeness: all questions posed by a narrative should be answered before the narrative ends, providing meaning to the spectator.

Furthermore, the notions outlined in the Interactivity Domain chapter proposed that interactivity involved in such a design endeavor should deploy not only the technological aspects of interactivity, but also provide room for narrative immersion to evolve. Following this stance, several elements were identified as essential, and therefore can be summarized to form a framework for an interactive design for this study:

1) Interface:
   - The design of the interface should be natural.
   - The design of the interface should be transparent.
   - The design of the interface should provide first-person activity.
2) Viewer oscillation:
   - The design should allow a switch between immediacy and hypermediacy.
   - The design of oscillation should provide more frequent interactive states than passive states.
3) Goal:
   - The interactive design should allow a user to move towards a narrative goal.

4) Information:
    - The extent of interactive and narrative information should not overwhelm the user.
5) Integration of actions:
    - Interactive design should integrate user actions that are non-trivial and contribute towards the development of the story.
    - Interactive design should deploy a clear role of the user's actions.
6) Interactive structure:
    - Interactive structure should maintain narrative coherence
    - Interactive structure should deploy productive and selective interactions.

The key points outlined above not only provide a summary of the main aspects of immersion, narrativity and interaction, but also constitute the theoretical framework of this study, that sets the criteria for the development of an interactive narrative. Moreover, they afford the main foundation for the research methodology that will be used for the evaluation of user experience in an interactive narrative.

By synthesizing various relevant theories from the reviewed literature, this theoretical framework thereby proposes that by following these established criteria, reconciliation between immersion and interactivity within interactive narratives should be possible. This allows postulating that if this framework is utilized when designing an interactive narrative, then narrative immersion and interactivity could be reconciled in one unified domain. Therefore, following such a stance, a hypothesis for this research can be stated as follows:

The lack of immersion in an interactive narrative is not inherent in interactivity per se, but it is a variable that can be modulated by certain design criteria.

Following this statement, the identification of such criteria should open the possibility for reconciling immersion with interactivity. Therefore, it can be implied that if this criteria, proposed by the theoretical framework is valid, then there should be no significant difference between achieving immersion in an interactive narrative and achieving immersion in a non-interactive narrative. Nevertheless, such a comparison is possible only if both are developed based on the same criteria for narrative immersion and narrative coherence and only an interactive narrative is deploying the design criteria for interactivity to compensate for the alleged expectancy in the decrease of immersion. This therefore poses that in order to evaluate the validity of this theoretical framework, two narrative forms- an interactive one and non-interactive one have to be developed following the proposed criteria, and later compared for apprehending how interactivity influences immersion in an interactive narrative. The following sections of this study therefore provide an insight on how this criterion was realized in the development phase of an interactive narrative, and later describe the experimental methodology used for the evaluation of the proposed theoretical framework.

IV. DESIGN OF AN INTERACTIVE NARRATIVE

Basing on the criteria proposed by the theoretical framework, this chapter provides an insight on the design of the interactive narrative for this study. The chapter discusses how the criteria proposed in the theoretical framework is adapted and realized in this project, and argues for the various choices in regards to narrative structure and interactive structure designs. Furthermore, the chapter describes the techniques that are utilized for maintaining narrative coherence and affording immersion within the realms of an interactive narrative. As the methodology for testing the above stated hypothesis requires two forms of narrative to be created, both of the forms are designed regarding the criteria set for the narrative structure and coherence, which are outlined in the first section of this chapter, while the second section of this chapter is only applied to the interactive form of the designed narrative.

1. Narrative Structure Design

The design of an interactive narrative begins with a choice of a type of story. According to Ryan's previously discussed arguments for each plot type in [33], an epistemic plot was chosen as the most appropriate for the design of an interactive narrative for this project. A mystery detective story, a standard representative of the epistemic plot, was selected, due to its compatibility with user's interactions and ability to incorporate the three types of narrative immersion, suggested by the theoretical framework. Moreover, this type of narrative, due to its inherit problem-solving scheme, promotes the systemic aspects of immersion in a sense that it deploys mental challenges of solving a mystery and interpreting given clues.

The designed narrative, tells a story of a respected detective investigating a murder of his ex-partner taking place in an isolated hotel on an island in the mid 1950's. The murder case is constituted by the events that took place in the past, and a mysterious murderer that aims to frame the detective for a crime he did not commit. The story unfolds by the user investigating the identity of the mystery killer, by sorting out the given controversial clues and making logical connections between them and the different suspects. Each step of the investigation triggers different events in the story and provides insights on the various suspects in the detective case, that influence the detective's judgment of who the real murderer is. The running time during the investigation, is the detective's enemy, as the murder case has to be solved before the next boat with the police arrives.

Regarding the described plot, the main characters in the narrative are the detective, the victim and the three possible suspects. Referring to the previously stated criteria, the characters are designed to have a mental life in a story that manifests in terms of their motivations for committing the crime. Basing on the settled criteria for emotional immersion, the characters are designed to be able to react to the environment by means of commenting on the protagonist's actions, and cueing him to follow the story in one way or the other. Moreover, each character is designed to have a clearly defined role and identity in terms of a story that remains stable throughout and until the end of the narrative. Nevertheless, to create suspense and involve spectator's intellectual properties into a narrative, the murderer identity is designed to be mysterious, but emerging through time of investigation. Furthermore, following the proposed criteria for creating emotional immersion, the narrative is designed in such a way that the spectator cannot avoid meeting the main characters of the story. While also, throughout time, the amount of information upon inner works and motivations of characters increases, allowing developing an involvement with the main characters of the story.

The narrative space is designed in a way that it would promote spectator's imagination, by not revealing all of the places of interest at once, by means of locked doors and "staff only" rooms. These then remain uncovered until a specific narrative event has occurred, or a certain clue in the investigation has been found. Additionally, the story environment is designed to promote the spectator's exploration, in terms of a detailed spatial design, and a deployment of hidden pathways and corners. On the other hand, the spatial setting remains constrained to the hotel boundaries, in order to limit the amount of the offered information and allow the spectator an easier apprehension of the environment. These design considerations closely coincide with the criteria settled for Spatial Immersion, in a sense that by deploying these elements the spectator should be involved in a mental construction of the topography of the story world.

The design of this detective mystery story implicitly incorporates temporal immersion and closely coincides with its specific elements proposed as criteria in the previous chapter. The story structure as such, deploys uncertainty in a way that until the final clues are gathered, the spectator is unable to know who the real murderer is. At the same time, suspense is created by presenting a murder scene in the beginning of the narrative, therefore affording temporal immersion throughout the entire narrative time. Additionally, the aspect of constantly elapsing time, is deployed as an element of suspense, in terms of a concrete time frame of the investigation, and putting the protagonist's life at risk, by means of making him as one of the suspects in the murder case. These design considerations closely coincide with the temporal aspects of immersion, and thereby should contribute towards sustaining the spectator's involvement with the narrative.

Referring back to Ryan's arguments in [2], a casual narrative structure type was argued to be the most applicable in an interactive narrative scenario. Our proposed design therefore deploys this narrative structure in a way that all the occurring events are followed by a causal chain in time and space which leads to a meaningful outcome. The narrative time is designed to run in a chronological consistency in order to ensure continuity between events and provide semantic relations between them. Therefore, the narrative is organized into clear time phases of beginning, development, complication, conflict, and conclusion. Here the beginning introduces the main characters, the environment and the purpose of the story, while the development phase provides an insight on the protagonist's and the suspects' characters. These are then followed by a complication – the time of the murder and its encounter, and later by the conflict phase, where the investigation is developed. All of the issues proposed by the narrative are at last resolved in a conclusion phase in terms of providing a clear ending, that manifests, only when the goal of the story has been achieved, and all questions posed by the narrative have been solved. As the narrative choice and structure design in this case are organized in a problem-solving scheme, the story explicitly possesses a clear goal – to solve a murder mystery case. To ensure the sense of completeness, therefore, the ending of the narrative is only available when all the events have been encountered and all the revealing clues have been gathered. This therefore guarantees the end of the narrative to serve as a conclusive stage, where all the questions posed by the story are answered.To inherit spatial-temporal variations into the plot line flashbacks are used. Nevertheless, they are designed not to interfere with the flow of the narrative as they are integrated with the events of the story only to remind the user of a particular cue or an object that has a significant meaning for the narrative to continue. Flash-forwards were omitted from the design considerations due to the possibility of ruining suspense in terms of knowing what is going to happen next.

2. Interactive Structure Design

To afford interactivity, the story is organized into an interactive structure (see figure 5), and is primarily based on the earlier discussed Hidden Story interaction model proposed by Ryan in [2]. As implied by the model, the story is designed in a way that the spectator is required to reconstruct a temporal sequence of events constituted by a story in order to unravel the hidden mystery. In our adapted version of the model the interactor is able to freely roam through the environment of the story world, producing non-linear pathways of the spatial investigation. Giving ability for the users to completely freely choose the narrative pathway in terms of investigation therefore supports the productive interactivity type. However, the overall narrative structure is kept linearly consistent, as every event can be triggered only in a chronological order independent of the spatial path of user's investigation. Meaning that events are regarded as conditional states that have to be met for the following events to continue, e.g. second event can only be experienced after experiencing the first one. This way, coherence, already inherit to a narrative structure should be ensured, and the flow towards the narrative goal should not be disturbed by the actions of the interactor.

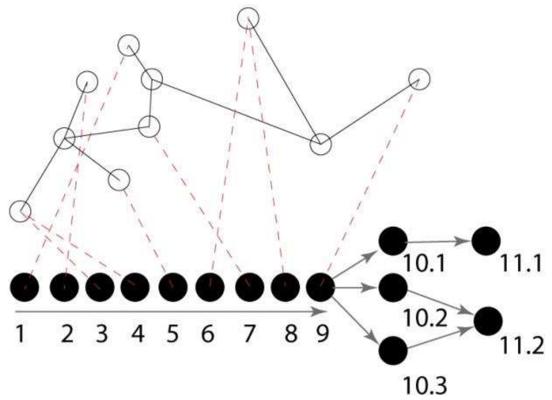

Figure 5 - Interactive Narrative Structure Design, horizontal axis – temporal sequence of events to be elucidated, vertical axis- spatial user's investigation

The adapted interactive structure also includes flashbacks to inherit spatial-temporal variations into the plot line. Such an adaptation, therefore inherits some of the properties of the Vector with Side Branches structure earlier defined by Ryan in [2]. As these temporal variations are implicit to the events of the narrative, they are deployed as unavoidable states, that provide story-related information for the user, rather than the states of choice that could potentially disturb coherence and the overall flow of the narrative.

Moreover, the previously settled criteria states that the interactive narrative design should embody non-trivial user interactions, giving ability for the users to choose the direction the narrative will take. To afford such a proposition, a branching storyline structure was incorporated at the end of the narrative sequence, allowing the interactor to choose the outcome of the story. The branching of the storyline therefore implies two possible endings of the narrative – one where the detective gets framed, in case he has made a wrong choice among the suspects, and the other one – where the detective catches the right murderer and solves the case. Nevertheless, narratively, both of the endings are identically important and neither constitutes a victory or a failure. This property of the interactive design structure therefore allows for selective interactivity to occur, putting the interactor in a role of an author by means of contributing towards the development of the story. Furthermore, non-trivial user's actions and a co-authorative story development are deployed in a sense that the overall interactive narrative structure provides room for mental interactivity to occur. This is intended through the design of controversial clues in an investigation, that are pointing to more than one suspect at a time, hence allowing the interactor to involve his intellectual thinking in solving the mystery.

Overall, the design of this interactive structure also closely coincides with the previously discussed design of the narrative structure. Because the structures implies chronological consistency among the events in the narrative, every experienced storyline, independent of the spatial pathway the spectator has taken, guarantees a logical and continuously corresponding temporal and spatial relationship between the experienced sequences of the story and thereby corresponds to the previously set criteria for the space and time continuum and maintains narrative coherence obtained with the design of the narrative structure.

As proposed previously, when design of an interactive narrative, the interface is essential, in a sense that it ties interactivity and narrativity together in one domain, and thereby contributes towards an immersive experience. Adapting the notions of the transparent and natural interface from Bolter and Grusin in [3], and Ryan in [33], the interface design discarded the use of menus, buttons or selection boxes. Instead, the design of an interface composed of symbolic performance of physical actions in a narrative environment. The choices of the user therefore consisted of walking through pathways of the virtual world and clicking on objects that serve as clues in the investigation. Nevertheless, in order to provide the user to easy orientation in a narrative environment and efficient apprehension of his actions, the designed interface has to provide some type of feedback. This was designed in terms of visual cues that should indicate the user's direction of movement, and an appearance of a picked up object in the screen. Moreover, the criteria for the design of the interface constituted that it should allow first-person activity. This was deployed in a sense that interactor is given a role of the main character, and plays the part of the detective in the story, and thus allows integrating user-actions in a story in a meaningful way. Moreover, the notion of the interactor as a protagonist coincides with the criteria settled for emotional immersion, and thereby should contribute towards the user's involvement with the story.

One of the last aspects settled in the criteria for the design of an interactive narrative, regarded user oscillation between the states of immediacy and hypermediacy, and hence interactivity and narrative immersion. The design of the interactive narrative deploys this in a sense that some of the events serve as linear narrative scenes that user has no control of. This furthermore promotes temporal immersion, as interactivity is limited at key points, and thus turns the avatar players into helpless spectators similar to those in film, and emotional immersion as that the reader is forced to face the destiny of a character without intervention. Nevertheless, following the criteria, the amount of these pre-defined linear narrative scenes is kept minimal, in a sense that interactive moments serve as the rule and passive moments as an the exception, which thereby allows the interactor to feel in control of the experience. Furthermore, the few linear scenarios are also used as elements for providing narrative completeness, since they both introduce and conclude the story, tying the user's interactive experience into a coherent structure.

V. IMPLEMENTATION OF AN INTERACTIVE NARRATIVE

In order to implement the above discussed design considerations, a choice of representational means had to be made. Therefore, the designed story was chosen to be implemented in a form of a 3D animation. Therefore, the entire environment and characters were built using a 3D content creation application Autodesk Maya and a game developer engine Unity3D. Animations used for simulating characters' movements and visualizing events of the story were created by using motion capture of a live actor. Actions were tracked by Arena OptiTrack motion capture system, composed by 16 infrared cameras that captured the body movements of an actor wearing a suit with reflective markers on the main joints.

To simulate the environment that supports the content of a mystery story, various visual and auditory elements had to be created. The overall environment was a hotel scene which included several locations within the hotel: the ceremony hall, the detective's room, the ex-partner's room, and the reception. There were hallways and stairs created for connecting different locations, which aimed to help subjects to explore the environment.

The visual elements had to be created with low resolution to improve the performance of the application on the mobile device. To create the illusion of high resolution several modelling and texturing techniques have been used, such as smoothing the edges of the models and adding bump maps.

In order to implement the notion of the interactor as a protagonist, first-person view of the camera was used. Nevertheless, for the previously outlined purposes of introducing the characters, pre-rendered animations were made utilizing third person view, giving a wider angle on the characters and the environment. In order to allow a better apprehension of the environment and to support the interactions, the users were provided a 360^{O} exploratory view in horizontal orientation and 120^{O} in vertical orientation.

Figure 6 – The interface of the developed interactive narrative

To implement a transparent interface, conventional touch-screen smart phone capabilities were deployed. For looking around in all directions, the sliding motions to the right, left, up, down, were used. Moreover, to implement the interactions with objects a two-time tap on the screen was utilized. In order to move around through space, a double tap on the floor was deployed. The feedback provided by the interface consists of a circular mark on a floor, indicating where the user's direction of movement. In order to provide feedback upon interaction with objects, a user was presented a virtual hand holding a specific object in the right corner of the screen.

Viewer oscillation was implemented in terms of switching between pre-rendered animation scenes and the interaction state. To ensure a high quality of the graphics and achieve more smooth animation patterns, the animation scenes were pre-rendered in Maya 3D, while all animations during the interaction state were imported into unity as real time animations. Similar techniques were utilized for creating flash-backs in the scene. Here the animations were pre-rendered in Maya 3D, and were given a blur effect in order to distinguish the temporal variations.

The audio elements used for the interactive narrative were recorded using a Shure SM-57 microphone through a Focusrite Saffire 6 sound card, using the integrated preamps. All voices were subsequently processed with equalising, compression, de-essing, saturation and limiting; Specific effects were applied in order to modify the characteristics of the voice (using effects such as Antares Mutator – emulating the modification of the length and properties of the throat, formant and pitch shifting) in order to give a specific „timbre" to the voice. Impulse responses from bakelite phones and old speakers were used for the telephone conversation and award speech, to make the voices appear coming from the specific place and device. Reverberation was subsequently added to all voices – in subtle amounts – to model the voices being in a room. The rest of the sounds were field recordings.

VI. METHOD OF RESEARCH

Various elements, constituted by the proposed theoretical framework, promoting an immersive but interactive experience, were attempted in the development of the interactive narrative for this study. In order to assess and evaluate the validity of this theoretical framework and to test the hypothesis of this research, a user study is therefore necessary to be conducted, with a main intent of testing the influence of interactivity on narrative immersion. Moreover, in order to evaluate the developed interactive narrative, a usability test has to be conducted for apprehending the status of the prototype and its applicability for the experiment. Therefore, firstly, this section of the study, will introduce the conducted usability test. Subsequently, a careful consideration regarding the choice of method for assessing immersive experiences for this study will be discussed, following by a detailed description of the experimental design and an explanation of the data gathering procedures of the experiment.

1. Usability Testing

Having the prototype of an interactive narrative developed, the first step into an evaluation process was to conduct a usability test with a main intent of assessing the overall status of the prototype and its applicability for the subsequent hypothesis testing. As the purpose of this test was to gather qualitative answers from the test subjects, interviews were chosen to be the most appropriate method for apprehending such data. Usability test questions can be found in the appendix 3.

The usability evaluation was conducted as an open-session interview, where the participants were asked to perform a talk-out-loud evaluation while testing the prototype. After the session, additional questions regarding the user's comprehension of the story, overall experience, the encountered issues and the status of the prototype were asked. In order to conduct the usability test, several participants from the Medialogy program at AAUC were involved. This testing sample was specifically chosen due to their overall high-level of understanding of the technological and contextual aspects implicit to this research. For future references and discussion upon the procedure and gathered results, the test session was video recorded. Relevant video material can be found on the DVD, attached with this paper.

From the evaluation session it was encountered that overall the participants enjoyed their experience, however, several issues regarding the usability of the prototype were pointed out. It was encountered, that most of the participants had a good apprehension of the story world, but however, had many difficulties using the interface to navigate through the environment. The main issue regarding their movement throughout the story world was due to lag. Moreover, some of the participants pointed out that the audio design interfered with their apprehension of the story, as due to excessive processing and improper level adjustments, the sounds were unclear and parts of the dialogue could not be understood. Further flaws were identified as lack of cues in sounds, especially when determining whether the detective was speaking or thinking, and also the absence of cues in flashbacks. Another shortcoming of the sound implementation was the lack of environmental sounds to coincide with the visual atmosphere, for example door locks, footsteps, and chatter in the background. Additionally, the variation of music in regards to the mood and the current situation of the detective should be implemented.

Moreover it was encountered that even though the participants were not able to approximate the original story in their retells, all of them felt narrative closure and were able to develop a story depending on the spatial investigation. Majority of them pointed out that few clues presented in the story were misleading and therefore should be readjusted in order to compensate for the lack of apprehension of the intended story.

These issues, especially due to their negative effect on the apprehension of the story, therefore proposed a questionable stance in terms of the applicability of the prototype for further evaluation of the hypothesis. As previously proposed, in order to test the hypothesis of this study, the immersive experience of the participant's in an interactive version and a non-interactive version of a narrative, designed following the same criteria for narrative structure and narrative coherence, had to be evaluated and compared to see whether or not the applied interactivity influences narrative immersion in an interactive narrative. However, regarding the gathered data from the usability test, it is evident, that such a comparison cannot be achieved, as the usability aspects and the flaws in the implementation could have an obvious negative effect on the overall evaluation of narrative immersion, and thereby deliver inconclusive and statistically invalid results. Due to the time constrains limiting the further development of the prototype, the conduction of the experiment for testing the hypothesis of this study therefore has to be postponed to after the submission of this paper. Nevertheless, various aspects, regarding the testing methodology and the experimental design remain essential for the purposes of this study, and can be developed for the future conduction of the experiment. The following sections of this chapter therefore present the possible testing methods that could be used for evaluating immersive experiences and narrative coherence, the experimental design considerations, and the possible techniques for gathering and evaluating such data in the future.

2. Measuring Immersion

The subjective nature and the diversity among the interpretations when conceptualizing immersion, consequently have led to vagueness among the proposals for measuring immersive experiences. As discussed previously, the main divergence in the academic research on immersion emerges between the underlying properties that define immersive experiences, as some identify the construct as an attribute of technology, while others acknowledge it as a deep state of psychological and cognitive involvement. Consequently, this discrepancy is also reflected in the proposed measurements for quantifying immersive experiences. Slater and Wilbur in [26], for instance, due to their technological apprehension of the concept, subsequently propose technological properties of a mediated environment as an objective measure for immersion. On the other hand, such scholars, as Witmer and Singer in [27], dismiss such definition and propose to measure immersion as a psychological phenomenon that leads to experiencing presence. Exemplifications of such discrepancy are plentiful, however, as our research proposed the view on immersion in regards to its narrative properties, and hence the viewer's psychological involvement with the story, immersion measurements only coinciding with such a perspective should be taken into consideration when investigating a suitable measurement method for quantifying immersive user experiences in the conducted experiment.

Referring to the criteria, previously settled by the theoretical framework in this paper, immersion in this study is regarded as a psychological experience of a greatly peculiar and individual nature. Due to this perspective regarding the phenomenon, subjective measures of immersion are therefore predominant in this specific case. Such subjective reports commonly manifest in the form of post-immersion questionnaires, think-out-loud or continuous subjective self-reporting tests. Nevertheless, the latter two, even though are not taking away much of the participants attention, might be likely to intrude their experience and therefore might affect the overall evaluation of immersion [32]. Post-session immersion questionnaires and interviews, though dependent on memory of the event, are considered to be more reliable and are proven to measure the proposed concept without directly interfering with the experience [32]. Nevertheless, even though, interviews, might offer more qualitative data, immersion questionnaires allow easier quantification and interpretation of the results [29], and thereby are considered to be more pertinent for gathering data in this study.

A relevant exemplification utilizing quantitative self-report metrics in a form of a post-session evaluation is an Immersive Experience Questionnaire (IEQ) by Jennett et al in [28], designed to measure the subjective levels of immersion. The proposed method evaluates user experience based on cognitive involvement, real world dissociation, challenge, emotional involvement and control factors, and thereby coincides with our previously proposed cognitive-psychological viewpoint on immersion. On the other hand, this method of measurement, does not allow enough detail to evaluate immersion in terms of its spatial, temporal and emotional properties, and thereby its applicability in a narrative scenario is questionable.

A questionnaire proposed by Green and Brock in [25], on the other hand, addresses immersion in terms of transportation, where contrary to the previous proposition of Jennett et al in [28], the intent is particularly set to measure user's absorption and transportation within narrative media. Despite the difference in terminology between immersion and transportation, the correlation between the concepts appears to be strong, and as discussed previously in the theoretical research chapter, the view on immersion in terms of transportation in the realms of this project is possible. Referring to the method proposed by Green and Brock in [25], the authors evaluate transportation by means of a Transportation Scale, consisting of questions regarding general immersion, and questions specific to a narrative included in the experiment [25]. The method was designed explicitly so it would cover the main dimensions of transportation, namely "emotional involvement in the story, cognitive attention to the story, feelings of suspense, lack of awareness of surroundings, and mental imagery" [25, p.703]. Prominently, as a number of correlates could be seen between these dimensions of transportation and the previously determined criteria for narrative immersion, this evaluation method coincides with the purposes of this study, and thereby can be utilized as a reference when constructing the questionnaire for this research.

While other methods of measuring immersion exist, namely those trying to objectify its implicit subjective nature, self-reported questionnaires, as the example proposed by Green and Brook, remain superior in terms assessing individual responses of participants. Alternative to subjective measurement approach, such propositions as gathering objective measures of immersion in terms of behavioral, physiological and performance measurements exist. Throughout a wide range of application areas, these methods, often regard immersion in terms of attention, reaction times and bodily changes [29]. From the reviewed literature it is evident, that not only none of these methods provide enough detail for grasping the essential aspects of narrative immersion, but also, many of them lack face validity and are still at exploratory research stages. Following these notions, the applicability of these measurements for this project is arguable, and therefore, further considerations upon such methods in this study are seen as irrelevant.

3. Measuring Narrative Coherence

Referring back to the criteria set in the theoretical framework of this study, narrative coherence was accentuated to be one of the most essential aspects contributing towards an immersive experience, and the one without which narrative immersion cannot exist. Following this postulation, it thereby is important to form an understanding of how such a phenomenon could be evaluated in terms of an interactive narrative.

While suggestions in measurements of narrative coherence are few, due to its implicit meaning, the construct is proposed to be measured in a form of evaluating a story retell [30], [31]. In these examples, evaluations concern either the approximation of the retell to the original story line or the participant's establishment of some coherent narrative meaning. An adaptation of such methodology seems relevant in our case, as it provides the means for quantifying narrative coherence, and thereby assessing user's narrative immersion. Nevertheless, a criterion for such an assessment needs to be settled, providing a method for distinguishing between the degrees of coherence. A noteworthy approach for such measurement is made by Hargood, Millard and Weal in [31], where narrative coherence is assessed through five variables: logical sense, theme, genre, narrator and style. The authors constitute these variables as the essential elements of narrative generation, and thereby propose the evaluation of narrative coherence through their identification and presence in a narrative. Following this notion, and the established criteria for narrative design in the theoretical framework of this study, we can postulate that narrative coherence can therefore be measured in terms of our proposed set of narrative elements. Referring to the theoretical framework, the proposed narrative elements were: Space, Time, Events, Characters, Context, and Narrative structure. Basing on the methodology in [31], we can then use the identification and presence of these elements encountered in the retells of the original story, to evaluate the degree of narrative coherence.

4. Testing Methodology

The immersion questionnaire comprised for this research uses a similar setup and measurement dimensions as proposed by Green and Brock in [25]. Nevertheless, as the questions proposed by the authors directly refer to the literary sense of reading narratives, a reformulation of the items into a context of interactive narratives is necessary. Moreover, to assess the spatial, temporal and emotional elements of immersion more directly, some additional questions, derived from the implicit characteristics of each immersive aspect have to be included. The final questionnaire for this research therefore consists of a total of thirteen questions, twelve for immersion and one for narrative coherence. The questions for immersion are categorized into four distinct categories of three questions each: spatial immersion, temporal immersion, emotional immersion, and general immersion. The first category consists of general questions regarding narrative immersion, where the overall participant involvement and concentration on the story is aimed to be measured (see questions 1- 3 in appendix2). Basing on the criteria constituted in the theoretical framework, the spatial immersion category is comprised of questions directly relating to the spectator's comprehension of, and involvement with the spatial setting of the story (see questions 4-6 in appendix2). The temporal immersion category consists of questions that address the extent, to which the spectators desire to know the causes and effects of central events of the story, and their curiosity of the different pathways the narrative could take (see questions 7-9 in appendix2). While the emotional immersion category is composed of questions that concern the extent, to which the experiment participants are involved with, and have comprehension of the roles of the characters in the narrative (see questions 10-12 in appendix2).

For an easier and more fluent analysis and interpretation of the responses, all the questions in the questionnaire, regarding immersion are presented in a form of a rating-scale. A Likert scale from 1 to 7 was chosen to be used, where 1 represented “not at all” and 7 - “very much”. A seven-point scale was selected among other possibilities, as it provides more resolution for answers than a five-point scale, but at the same time still allows an easy understanding of what each rating represents. Moreover, like other odd-numbered scales, it provides the middle rating, which allows a neutral score if a subject is undecided. Nevertheless, when evaluating data, gathered from the rating scales, it is noteworthy to keep the central-tendency bias in mind.

Evaluation methodology for narrative coherence is based on the testing methods discussed in the previous section of this chapter. Following, a task was created, asking the participants of the experiment to write a retell of the experienced story (see question 17 in appendix2). The retells should then be evaluated, depending on their juxtaposition with the original story line, and the degree to which the basic narrative elements can be identified and are present. To quantify the gathered data, an identical Likert scale is used, 1 being non-coherent, while 7-being very coherent. The task is presented in a form of an open-ended question in the questionnaire.

Moreover, regarding our own personal interests, several questions regarding the overall experience were added into the questionnaire, regarding the usability, the audio and the visual aspects (see questions 13 -16 in appendix2). Nevertheless, these will not be utilized when calculating the results for immersion nor for narrative coherence. For a more detailed overview of the testing material, the full version of the questionnaire, comprised of both, the narrative immersion and narrative coherence sections can be found in the appendix 2 of this paper.

5. Experimental Design and Procedure

In order to assess the validity of the proposed theoretical framework and to apprehend whether or not interactivity affects narrative immersion, the designed experiment should consist of evaluating and comparing the user experience in interactive and non-interactive scenarios. Therefore, the research experiment is designed to have two conditions, one for each scenario. Throughout further discussions, presented in this paper, the interactive condition will be termed as Test Condition1 and non-interactive as Test Condition2. For this purpose, two forms of a narrative following the same story were created: a linear form, presenting one of the possible linear story lines from the interactive scenario, and an interactive form, involving user participation. Given, that interactivity is objectively implicit only in the Test Condition1, and that the criteria for immersion, narrative structures, and narrative coherence, presented within the theoretical framework have been met in the design of both conditions, a comparison of the immersive capabilities between the two can then be possible. To perform the comparison, in both tests, the participants should be presented the same immersion questionnaire and an identical task for apprehending data on narrative coherence. The presentation of identical test methods for both groups thereby should provide an apprehension, of if, and how immersive experience of the participants differs when comparing the both conditions. In case that the hypothesis of this research has been accepted, the comparison of two conditions should then propose no significant difference between the results of the two tests, or should have an increased tendency in the Test Condition1, which then would validate the proposed theoretical framework and hence imply that interactivity did not have a negative effect on immersion, and that decrease in immersion is not implicit to interactivity per se.

As the test results regarding immersion, and especially, narrative coherence, might be influenced by repeated viewings of the narrative, all the subjects should be constrained to participating in the experience only once. In case an additional viewing would be insisted, a possibility of a repeated viewing can only permitted after the evaluation of the experience, and should not be included in the overall evaluation of the experiment. Basing on these reasons, for the validity purposes of the comparison between tests a between-subjects test design is chosen to be used, implying that each participant can be exposed to only one testing condition.

As the objective of this study is not to describe a population, but to study the effects of one variable on another, and therefore, both probability and non-probability sampling techniques can be considered to be applicable for the purposes of this research. And nevertheless, even though the non-probability sampling method is more accessible and possible in the realms of this study, when compared to the probability method, the latter one could be advantageous for making unbiased estimates of the population totals. Moreover, in order to gather statistically significant results, a sample size of at least 20-30 participants should be used.

In order to reduce bias that might arise due to preconceived notions of the experiment, prior to the test, all of the participants should be given the same short introduction to the experimental procedure, with no mention of immersion or narrative coherence. At the beginning of the experiment, test subjects should also be introduced to the interface and in case of the interactive testing condition, the controls of the narrative. After the experiment, the participants should be asked to fill out the questionnaire and to perform a story retell task, as outlined earlier in the testing methods chapter. To ensure a more efficient handling of data, the post-session evaluation could be carried out in a digital form.

VII. RESULTS

In this study the interest was set on apprehending how interactivity influences narrative immersion in an interactive narrative. The hypothesis previously postulated that by utilizing the design criteria, proposed by the theoretical framework of this study, reconciliation between immersion and interactivity in an interactive narrative would be possible. As outlined throughout the methodology chapter, the hypothesis of this study should be tested by evaluating and comparing immersion and narrative coherence scores gathered from the two testing conditions, namely Test Condition1 and Test Condition2. Where the criteria for interactivity is objectively implicit only in the Test Condition1, and that the criteria for immersion, narrative structures, and narrative coherence, presented within the theoretical framework is met in the design of both conditions. Moreover, it was argued that, in order for narrative immersion to exist, narrative has to provide a coherent meaning to the participant. This was suggested to be done by evaluating the gathered retells of the narratives from the questionnaire, and rating them according to the extent the narrative elements were present and could be identified in the participant's approximations of the story.

In order to determine how immersed the participants felt with the narrative, the average immersion values for each participant should be calculated by taking the mean value of all the answers from the 12 questions regarding narrative immersion in the questionnaire for both testing conditions separately. In order to evaluate whether there is a significant difference when comparing narrative immersion between the two testing conditions, a statistical hypothesis test can be made by utilizing the two-tailed T-Test method. This method would therefore provide us with a p value, indicating the rejection (if p is less than the alpha value of 0.05) or acceptance (if p is bigger than the alpha value of 0.05) of the null hypothesis. This would hence prove or disprove our proposed hypothesis for this study.

As stated previously, in order to apprehend the different types of immersion, namely emotional, temporal, and spatial, the questionnaire consists of three categories, one for each type of immersion. In order to evaluate whether there is a significant difference when comparing each type of immersion between the two testing conditions, an identical T-Test method could be used. This implies that in order to determine how emotionally, spatially or temporally immersed the participants felt with the narrative, the average immersion values for each immersion category have to be calculated separately by taking the mean value of the scores from the three questions of one specific category. Therefore, an average score for all the participants for each type of immersion could be calculated in one testing condition and compared to the average score of the corresponding immersion category in another condition. The comparison could be done by performing a T-Test, to check for statistically significant differences separately in temporal, spatial and emotional immersion categories between the results of the two testing conditions. By performing such a test, we would then be able to apprehend if, and how each of the immersion aspects was influenced when applying interactivity. This data would furthermore be useful to discuss and evaluate the results gathered for the overall narrative immersion, in terms that the overall narrative immersion scores could be influenced by some of the immersion factors more than the others.

Referring to the statements made earlier in this study, narrative coherence was argued to be one of the most crucial aspects for narrative immersion to exist. In the testing methodology, it was proposed to test this aspect by evaluating story retells provided by the participants, rating them on a scale from 1 to 7, depending on their proximity with the original story line, and the degree to which the basic narrative elements can be identified and are present. In order to test whether or not narrative coherence was influenced by interactivity an identical methodology of utilizing a T-Test could be used. Moreover, to test the relationship between narrative immersion and narrative coherence, Pearson Correlation can be used to find the correlation coefficient between narrative immersion scores and narrative coherence scores in both testing conditions. The correlation coefficient should indicate the degree of association between the two constructs, where scores from -1 to -0.3 (strong to weak correlation) would indicate a negative association (the greater the immersion, the lower the narrative coherence), and +0.3 to +1 (weak to strong) would indicate a positive association (the greater the immersion, the greater the narrative coherence).

VIII. DISCUSSION

Even though, depending on the state of the research it is not possible to deliberate upon the gathered results and the validity of the theoretical framework developed in this study, it is however necessary to discuss the overall status of the prototype and the following phases of this study. Firstly, in order to conduct a valid experiment, the issues encountered through a usability test have to be overcome through a second prototype development phase.

As outlined earlier, the main flaws exist in terms of usability of the overall experience, and the sound design. In order to overcome these flaws in the next prototype stage, therefore more considerate choices when implementing these outlined aspects have to be made. In order to resolve the issues regarding the sounds, firstly a more clear structure of the sound story-line has to be made, as it is crucial for a proper apprehension of the narrative, and hence affects the overall narrative coherence. Moreover, these narratively essential audio elements have to be redesigned with proper processing and level adjustments in order to achieve more clarity and understanding of the dialogues and monologues in the story.

Concerning the usability of the interface, the main issues regarded the user's movement throughout the environment and therefore should be resolved with the second stage of the prototype. The lag issue should be resolved by looking at different rendering techniques. Additionally, to get a better understanding of the intended story the clues have to be further emphasized.

Subsequently to the development of the second prototype, another usability test has to be conducted in order to ensure that the issues postulated above are solved and would not interfere with the conduction of the experiment for testing the hypothesis of the study. After all the issues have been resolved and a testable prototype has been developed, the experiment of this study will be conducted, utilizing the previously outlined testing methodology and the experimental design setup. Following these considerations, therefore should allow to test the validity of the developed theoretical framework, and to apprehend the influence of interactivity on immersion in an interactive narrative. The second phase of the prototype, the test procedure and the gathered findings will be presented at the following project exam session at AAUC.

IX. CONCLUSIONS

Throughout this study, various aspects of interactivity and immersion in the realms of an interactive narrative have been outlined. The findings obtained through the investigation have led to deliberate conceptualizations of immersion, narratives and interactivity, and have provided an apprehension of their implicit fundamental elements and their convergence in the realms of interactive storytelling. The distinction of these elements furthermore supported an establishment of a fundamental theoretical framework that set the criteria for the design of an interactive narrative, with a main intent of reconciling immersion and interactivity in one unified domain. Following these design criteria, an interactive narrative was designed and implemented for the subsequent purposes of evaluating an immersive experience.

Nevertheless, the paper only delves into the theoretical assumptions and propositions, and thus the methodologies and hypothesizes presented throughout should be verified in terms of a scientific experiment for concrete conclusions to be drawn. Nevertheless, this investigation affords a solid foundation for continuing the research in the future and can be utilized as a theoretical guideline for related projects in the fields of interactive narratives and immersion.

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APPENDICES

I. INTERACTIVE NARRATIVE STRUCTURE

The graph bellow represents a complete version of an interactive narrative structure used for the development of the interactive narrative for this project.

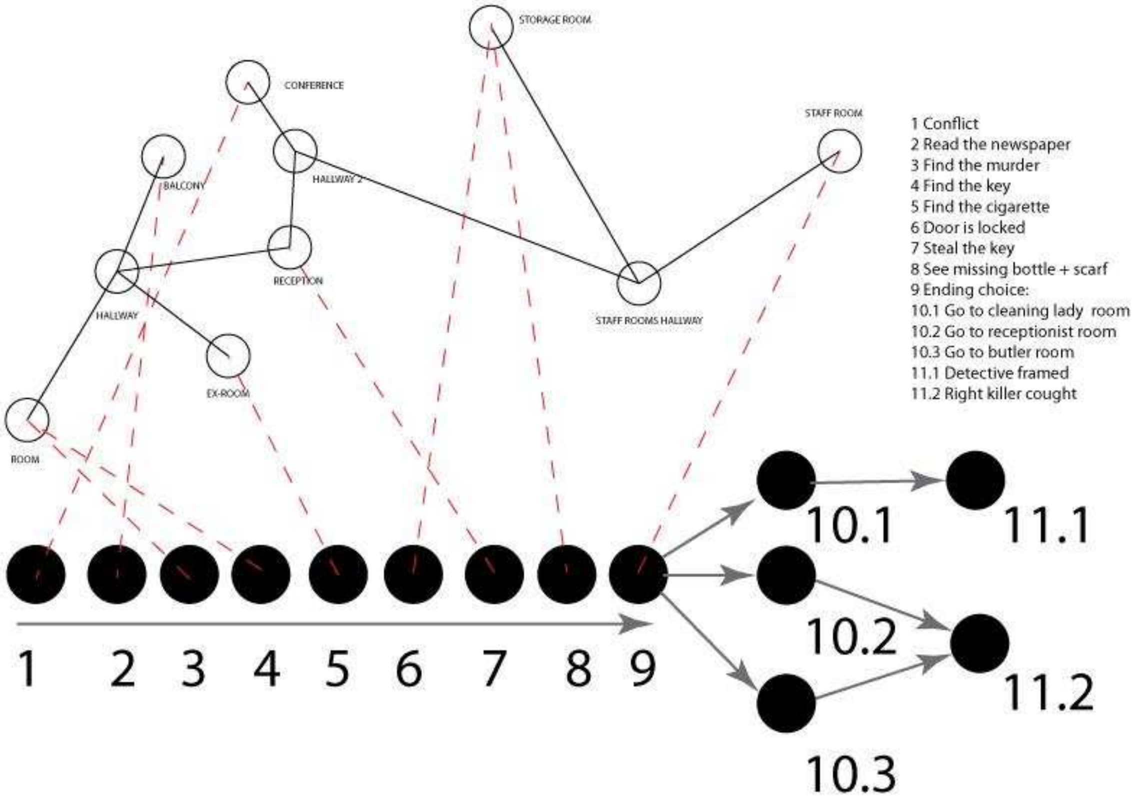

Figure 7 - Interactive Narrative Structure Design, horizontal axis – temporal sequence of events to be elucidated, vertical axis- spatial user's investigation. Black circles- main events in the story, White circles – all possible places in the space of the story for the events to take place, Arrows and numbers – the sequential order of the investigation.

II. QUESTIONNAIRE

All questions contained in this questionnaire are confidential and will be used only for the purposes of evaluating and discussing this project.
We kindly ask you to remain discrete about the test procedure and the experiment throughout the remains of the day.
THANK YOU!

* Required

1. During the experience, I was able to easily understand what was happening in front of me. *

	1	2	3	4	5	6	7	
not at all	◯	◯	◯	◯	◯	◯	◯	very much

2. To what extent, did I feel distracted by the things happening outside the story world *

	1	2	3	4	5	6	7	
not at all	◯	◯	◯	◯	◯	◯	◯	very much

3. To what extent, was I able to concentrate on the story *

	1	2	3	4	5	6	7	
not at all	◯	◯	◯	◯	◯	◯	◯	very much

4. The space presented, encouraged me to explore and imagine the environment that was beyond the doors and stairs of the story world *

	1	2	3	4	5	6	7	
not at all	◯	◯	◯	◯	◯	◯	◯	very much

5. I found it easy to orient myself within the space described by the story. *

	1	2	3	4	5	6	7	
not at all	◯	◯	◯	◯	◯	◯	◯	very much

6. I could easily understand the overall space described by the story. *

	1	2	3	4	5	6	7	
not at all	◯	◯	◯	◯	◯	◯	◯	very much

7. During the experiment, I was curious to find out how the story would end. *

	1	2	3	4	5	6	7	
not at all	◯	◯	◯	◯	◯	◯	◯	very much

8. During the experiment, I found myself speculating how the choices of the main character could influence the story. *

	1	2	3	4	5	6	7	
not at all	◯	◯	◯	◯	◯	◯	◯	very much

9. Among the different characters, I found myself wondering who the real murderer in the case was. *

	1	2	3	4	5	6	7	
not at all	◯	◯	◯	◯	◯	◯	◯	very much

10. While I was experiencing the story, I could easily understand the role of the main character. *

	1	2	3	4	5	6	7	
not at all	○	○	○	○	○	○	○	very much

11. While I was experiencing the story, I was interested in the fate of the main character. *

	1	2	3	4	5	6	7	
not at all	○	○	○	○	○	○	○	very much

12. I had a good understanding of the main characters in the story. *

	1	2	3	4	5	6	7	
not at all	○	○	○	○	○	○	○	very much

13. To what extent the interactions provided by the interface seemed natural to me *

	1	2	3	4	5	6	7	
not at all	○	○	○	○	○	○	○	very much

14. To what extent did the visual elements involve me in the experience *

	1	2	3	4	5	6	7	
not at all	○	○	○	○	○	○	○	very much

15. To what extent did the audio elements involve me in the experience *

	1	2	3	4	5	6	7	
not at all	○	○	○	○	○	○	○	very much

16. To what extent did I feel it was easy to use the application *

	1	2	3	4	5	6	7	
not at all	○	○	○	○	○	○	○	very much

17. How would you retell the experienced story to a friend? *

Please describe your version of the experienced story bellow, using as much detail as possible and involving considerations upon such aspects as characters, events, space, time, conflicts, motivations, ending and etc.

Submit

III. USABILITY INTERVIEW QUESTIONS

1. Please think out loud when using the application.
2. Can you please retell the story?
3. What was your role in the story?
4. Who is the murdered guy?
5. Who was the murderer in the story?
6. How much do you think the usability of the application affected your understanding of the story?
7. Please point out the issues that you have experienced during the test.